

# Amortized Bayesian parameter recovery for dynamical systems with conditional flow matching

The `amortix` package: architecture, evaluation methodology, and benchmarks

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## Abstract

We describe `amortix`, a package for amortized Bayesian estimation of the parameters of dynamical systems — ordinary and stochastic differential equations and discretized PDEs — from noisy observations at a finite set of time points. A conditional flow-matching model is trained once on simulated (parameter, data) pairs and thereafter produces posterior samples for any new observation set without likelihood evaluations or per-instance optimization; a single trained network handles any number and placement of observation points, the setting required for sequential experiment refinement and Bayesian experimental design. The package couples three components that we argue are only useful together: (i) a set-transformer encoder whose input representation is learnable rather than prescribed — a monotone mixture-of-scales reparametrization of the value axis, attention phases computed from physical observation times, and explicit set-cardinality features; (ii) a benchmark suite of fifteen systems spanning finance, neuroscience, epidemiology, ecology, pharmacokinetics, classical mechanics, and reaction–diffusion PDEs; and (iii) an evaluation harness in which simulation-based calibration is used only as a screen, and the instrument of record is comparison against exact or validated Markov chain Monte Carlo reference posteriors on the same observed points. Trained once per system in this design-amortized mode, the default network matches exact or validated reference posteriors to median Fréchet distances (in reference-normalized coordinates) of 0.01–0.07 on four of six reference-equipped systems — evaluated across design sizes with a fresh random design per test observation set, where it is comparable to or better than networks trained for each single design — while the two elevated entries are measured effects: 0.19 on a five-parameter jump-diffusion that is capacity-limited at a fixed simulation budget, and 0.18 on dense-design pharmacokinetics, which meets an amortized precision floor of  $\sim 10^{-3}$  of the prior range that we measure directly; 34 of 36 parameters across the nine systems without a tractable reference pass simulation-based calibration, and residual errors decay steadily with the simulation budget. We compare against neural posterior estimation and flow-matching posterior estimation from the `sbi` package, both network families of BayesFlow 2, and a port of the Simformer, on three systems with exact or validated references. On a multiplicative process, all external configurations either inflate the volatility posterior by factors of 1.4–3.2 or become overconfident (widths 0.4–0.9) depending on the input coordinates they are given, while the learnable representation recovers the exact posterior from raw data. On an additive control process the same external tools improve markedly — the systematic bias disappears and the width inflation shrinks from 2.5–3.2 to 1.6–1.9 — which localizes the dominant failure to input conditioning while showing that a residual representation gap remains even in the benign case. We document this failure mechanism — the conditioning of the encoder input — in three distinct forms, together with learnable modules that repair each one.

## 1 Introduction

Recovering the parameters of a differential-equation model from noisy measurements of its solution is one of the basic inverse problems of applied science: a modeler writes down a dynamical system — an ODE, an SDE, or a PDE — whose right-hand side depends on unknown physical constants, observes a trajectory at a finite set of time points, and asks what the observations say about the constants. Like most inverse problems, this one is ill-posed in the practical sense: different parameter values can produce nearly indistinguishable observations, especially when the state is partially observed or the sampling is sparse, so a point estimate both hides the ambiguity and is unstable under noise. The Bayesian formulation

resolves this by reporting the full posterior distribution over parameters given the data — ambiguity becomes posterior spread and correlation, quantities one can act on — at the price of two computational obstacles. First, for most stochastic and partially observed systems, the likelihood of the observations requires integrating over unobserved trajectory segments and is intractable or expensive. Second, modern workflows need posteriors *repeatedly*: calibration studies process thousands of synthetic observation sets, sequential experiments update the posterior after every new measurement, and Bayesian experimental design compares candidate observation schedules by the posteriors they would produce — so per-instance MCMC or optimization inside such loops is the dominant cost even when a likelihood is available.

Simulation-based inference (SBI) addresses both issues by learning a conditional generative model of the parameters given the data from simulated pairs — parameters drawn from the prior, observation sets produced by the simulator — thereby amortizing the cost of inference into a one-time training phase (Cranmer et al., 2020). Neural posterior estimation (NPE) and its sequential variants (Papamakarios and Murray, 2016; Greenberg et al., 2019) fit normalizing flows by maximum likelihood; flow-matching and diffusion variants (Wildberger et al., 2023; Gloeckler et al., 2024) replace the density estimator with transport-based generative models. Mature software exists: the **sbi** toolkit (Boelts et al., 2024), BayesFlow (Radev et al., 2020), and the **sbibm** benchmark collection (Lueckmann et al., 2021).

This report describes **amortix**, a compact PyTorch package (about 3,500 lines; one default model of  $4.8 \times 10^5$  weights) for amortized Bayesian parameter recovery in dynamical systems. Three aspects of its design distinguish it from the toolkits above, and the report is organized around measuring their consequences.

The first concerns the interface between raw data and the network. Conditional density estimators are conventionally fitted to a fixed-length data vector after global standardization; the estimator sees whatever coordinates the user chose. We show in Sec. 8.3 that on multiplicative and heavy-tailed processes this convention fails in a specific, reproducible way across four different generative engines from three codebases, and that supplying hand-derived coordinates does not repair it uniformly: with identical inputs, different implementations end up over-dispersed, overconfident, or biased. In **amortix** the transformation from observations to network inputs is itself part of the trained model — a monotone mixture-of-scales reparametrization of the value axis, observation times entering attention as continuous phases, and explicit design-size features (Sec. 4); each component is motivated by a measured failure and validated by an ablation.

The second concerns observation designs. **amortix** trains a single network to approximate  $p(m \mid S)$  coherently for every subset  $S$  of the measurement points of an experiment — arbitrary sizes and placements, including spatio-temporal sensor designs for PDEs (Sec. 8.5). This is the object consumed by sequential refinement and Bayesian experimental design, and supporting it changes the training procedure (Sec. 5): designs must be resampled fresh at every optimizer step, and the distribution of design sizes becomes a modeling choice with measurable consequences.

The third concerns evaluation. Simulation-based calibration (Talts et al., 2018) is a necessary-condition test whose statistical power at practical sample sizes is limited; we measure that limit (Sec. 6) and use SBC only as a screen. The instrument of record throughout is the comparison of posterior mean and width against an exact or validated-MCMC reference posterior on the same observed points, and the package ships this machinery — closed forms, exact-likelihood adaptive Metropolis samplers, chain validation — as part of its public API. The concern motivating this discipline is documented in the SBI literature: learned posteriors can be confidently wrong in ways that rank diagnostics do not detect (Hermans et al., 2022).

The remainder of the report presents the method (Sec. 3), the architecture (Sec. 4), training procedures (Sec. 5), the evaluation methodology (Sec. 6), the benchmark suite (Sec. 7), experimental results including comparisons with **sbi**, BayesFlow, and the Simformer on three systems with exact or validated references (Sec. 8), an analysis of the failure modes encountered during development (Sec. 9), and limitations (Sec. 10). Appendix A defines every benchmark system with its equations, priors, and observation model; Appendix C details the reference posteriors; Appendix D maps the terminology of this report to package options.

## 2 Problem formulation

Let a dynamical system depend on a parameter vector  $m \in \mathbb{R}^d$  with a known prior  $p(m)$ , uniform on a box in all benchmarks below. The system produces a state trajectory  $x(t)$  — through an ODE  $\dot{x} = f(x; m)$ , an SDE  $dx = f(x; m) dt + \sigma(x; m) dW_t$ , or a spatially discretized PDE — and the trajectory is observed

through a noisy channel at  $K$  points:

$$y_i = h(x(t_i)) + \varepsilon_i, \quad \mathcal{D} = \{(t_i, y_i)\}_{i=1}^K. \quad (1)$$

We call  $\mathcal{D}$  the *observation set* of one experiment; it is the entire input to inference. Training and evaluation below involve many independently simulated observation sets, each paired with the parameter that generated it. The observation operator  $h$  may expose only part of the state: in the benchmarks below, only the price of an asset while its stochastic volatility is latent, only the membrane voltage of a neuron while its gating variables are latent, or only three spatial sensors of a PDE field. The target is the posterior  $p(m \mid \mathcal{D}) \propto p(\mathcal{D} \mid m)p(m)$ .

Three standing assumptions delimit the setting. The parameter is finite- and low-dimensional ( $d = 2$ – $5$  in every benchmark below) with a prior that factorizes across components and has tractable marginal CDFs (uniform boxes throughout; the construction extends to any such prior). The simulator can be run forward cheaply and repeatedly, but nothing else is required of it — no likelihood evaluations, no gradients. Observation times (and sensor indices) are known exactly; the measurement noise  $\varepsilon_i$  is independent across observations with a known per-system law.

**Running example.** The Ornstein–Uhlenbeck process is used throughout to make constructions concrete:  $dX = -\theta X dt + \sigma dW$  with  $m = (\theta, \sigma)$ , prior box  $[0.3, 3] \times [0.2, 1.5]$ , stationary start  $X_0 \sim \mathcal{N}(0, \sigma^2/2\theta)$ . An observation set is, for instance,  $K = 74$  noisy values of  $X$  on a fixed two-rate grid, or — in the variable-design setting — any  $K$  values at arbitrary times. Its posterior has the features that recur across the suite: the diffusion coefficient  $\sigma$  is identified sharply by high-frequency increments; the reversion rate  $\theta$  only weakly over a finite horizon; and the width of both marginals must shrink as observations accumulate. An estimator can therefore be wrong in the posterior mean, in the posterior width, or in the width-versus- $K$  trend, and the evaluation methodology of Sec. 6 is built to detect all three.

Although the posterior itself is intractable, sampling from the *joint* distribution of parameters and data is easy: draw  $m_j \sim p(m)$ , simulate the system, apply the observation channel. This yields arbitrarily many training pairs  $(m_j, \mathcal{D}_j)$ , and a conditional generative model trained on them approximates the posterior for every observation set simultaneously — the cost of inference is moved into a one-time training phase, and inference for a new observation set is a forward pass. We impose one requirement beyond standard amortization: a single trained network must remain correct for any number  $K$  and any placement of observation times (and, for PDEs, sensor positions). Formally, for every subset  $S$  of the measurement points of one experiment the network must approximate  $p(m \mid S)$ , and the family of answers must be coherent — in particular the posterior must widen as points are removed. We refer to this requirement as *design amortization*; it is the object consumed by sequential refinement and by Bayesian experimental design, where candidate observation schedules are ranked by the posteriors they would produce.

### 3 Posterior sampling by conditional flow matching

The idea of the method is to represent the posterior not by a density formula but by a *transport*: a learned, data-conditioned deformation that carries a fixed reference distribution onto the posterior. Sampling then amounts to drawing from the reference and applying the deformation — there is no normalizing constant to compute and no invertibility constraint to build into the network, because the deformation is defined implicitly as the flow of a velocity field, and the field itself is fitted by ordinary least-squares regression on simulated pairs. This construction is conditional flow matching (Lipman et al., 2023).

Concretely, parameters are first reparametrized componentwise,

$$m \mapsto \Phi^{-1}(F_{\text{prior}}(m)) \in \mathbb{R}^d, \quad (2)$$

where  $F_{\text{prior}}$  collects the marginal prior CDFs and  $\Phi$  is the standard normal CDF. In the new coordinates the prior is exactly  $\mathcal{N}(0, I)$  and box constraints disappear; usefully, the uninformative limit then costs nothing — when the data carry no information the posterior coincides with the reference and the transport is the identity. From here on we work exclusively in these coordinates and keep writing  $m$  for the parameter in them; the inverse of (2) is applied exactly once, at the end of sampling, to return draws on the original scale.

The transport is indexed by a *virtual time*  $\tau \in [0, 1]$  — an algorithmic variable with no relation to the physical time  $t$  at which the dynamical system is observed;  $t$  appears only inside observation sets,  $\tau$  only in the transport. Subscripts in  $\tau$  denote the position along the transport path:  $m_1$  is a parameter

drawn jointly with its observation set  $\mathcal{D}$  (the endpoint),  $m_0 \sim \mathcal{N}(0, I)$  is pure reference noise — not the transform of any parameter, but the raw material that the transport will have turned *into* a posterior draw by  $\tau = 1$  — and between them lies the training interpolation  $m_\tau = (1 - \tau)m_0 + \tau m_1$ . The velocity network is trained to predict the direction of this path by regression:

$$\mathcal{L}(\theta) = \mathbb{E}_{m, \mathcal{D}, m_0 \sim \mathcal{N}(0, I), \tau \sim U[0, 1]} \|v_\theta(m_\tau, \tau, \mathcal{D}) - (m_1 - m_0)\|^2. \quad (3)$$

The minimizer of (3) is the marginal velocity field whose flow transports  $\mathcal{N}(0, I)$  at  $\tau = 0$  to  $p(m_1 | \mathcal{D})$  at  $\tau = 1$  (Lipman et al., 2023). Sampling draws  $m_0 \sim \mathcal{N}(0, I)$ , integrates  $dm/d\tau = v_\theta(m, \tau, \mathcal{D})$  from  $\tau = 0$  to 1, and maps the endpoint back through the inverse of (2); the numerical settings of the integration are implementation details, collected in Sec. 5. The same estimator family, with a different conditioning architecture, is available in `sbi` as FMPE (Wildberger et al., 2023); the comparison of Sec. 8.3 therefore isolates the effect of the input representation from the choice of generative engine.

No likelihood evaluations, simulator gradients, or per-instance optimization are required at any stage; the simulator is executed only forward, during training-set generation.

The training objective is consistent in the following sense.

**Proposition 1** (consistency). *Let  $(m, \mathcal{D}) \sim p(m)p(\mathcal{D} | m)$  and let  $m_1$  be given by (2). If the loss (3) is minimized over all measurable velocity fields, then for  $p(\mathcal{D})$ -almost every  $\mathcal{D}$  the flow of the minimizer transports  $\mathcal{N}(0, I)$  at  $t = 0$  to the conditional law  $p(m_1 | \mathcal{D})$  at  $t = 1$ ; equivalently, the sampling procedure above draws exactly from the posterior (Lipman et al., 2023; Wildberger et al., 2023).*

The proposition is an idealization: with a finite network, a finite simulation budget, and a numerical solver, no calibration guarantee survives, and the residual error becomes an empirical quantity. Section 6 builds the instruments that measure it.

## 4 Conditioning architecture

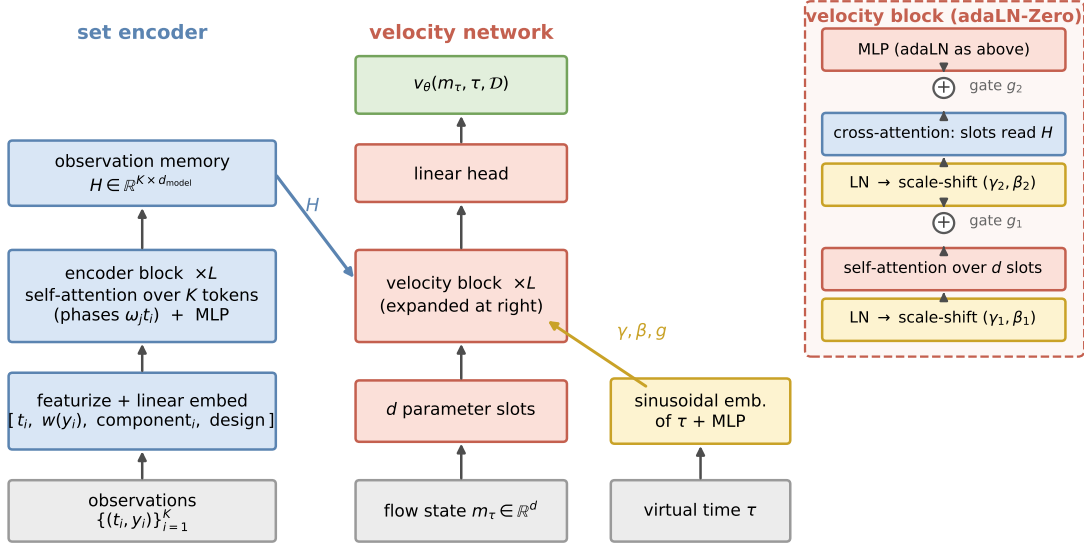
The velocity field  $v_\theta(m, \tau, \mathcal{D})$  must be conditioned on an observation set of time-stamped measurements, and three properties of that set drive the architecture: the number of observations  $K$  varies from experiment to experiment (by two orders of magnitude in the variable-design setting); the observation times are arbitrary real numbers, in general not equally spaced, so positional schemes built for uniform sequences do not apply; and the times carry all ordering information, so the output must not depend on how the observations happen to be stored. These properties rule out the fixed-length-vector conditioners that are the default in NPE implementations, and they are naturally met by a transformer operating on one feature vector per observation — throughout this section, a *token*.

### 4.1 Overall scheme

The network evaluates  $v_\theta(m, \tau, \mathcal{D})$ , where  $m$  is the normalized parameter vector partway along its transport path — the interpolation  $m_\tau$  at training time, the current ODE state at sampling time — and  $\tau$  is the virtual time of the transport. The design follows the encoder–decoder pattern that is standard in conditional flow-matching and diffusion models: the data are encoded once, and a decoder combines that encoding with the object being generated,

$$v_\theta(m, \tau, \mathcal{D}) = \text{Dec}(m, \tau, \text{Enc}(\mathcal{D})), \quad (4)$$

with explicit types: `Enc` maps the observation set — the  $K$  time-stamped measurements of one experiment — to a memory of  $K$  vectors,  $\text{Enc}(\mathcal{D}) \in \mathbb{R}^{K \times d_{\text{model}}}$ , computed once per observation set, and `Dec` :  $\mathbb{R}^d \times [0, 1] \times \mathbb{R}^{K \times d_{\text{model}}} \rightarrow \mathbb{R}^d$  combines the current transport point and the virtual time with that memory. Two differences from the image and text instances of this pattern dictate the concrete form (Fig. 1). First, the conditioning is not a fixed-size embedding but an observation set whose size  $K$  varies: `Enc` is therefore a transformer over per-observation tokens, its output is a *memory* of  $K$  encoded states rather than one pooled vector, and `Dec` reads the memory by cross-attention — the one standard mechanism indifferent to  $K$ . Second, the generated object is not an image but a short vector of  $d$  parameters, so the decoder is small: the  $d$  components of  $m$  are treated as  $d$  tokens, self-attention among them expresses posterior correlations (the ridge of the Fisher–KPP posterior, the drift–volatility coupling of GBM), and the virtual time  $\tau$  enters every block through zero-initialized adaptive layer normalization (adaLN-Zero), the conditioning mechanism of diffusion transformers (Peebles and Xie, 2023), so that the untrained network is an identity-like map.



**Figure 1:** Model architecture; arrows denote data flow, bottom to top. Left: each observation is featurized and linearly embedded into one token;  $L$  transformer encoder blocks, whose attention phases are computed from the physical observation times (Sec. 4.2), produce a memory  $H$  of  $K$  encoded states. Middle: the  $d$  components of the flow state  $m_\tau$  form  $d$  parameter-slot tokens processed by  $L$  velocity blocks and a linear head into the velocity  $v_\theta$ . Right: one velocity block expanded — self-attention across parameter slots, cross-attention reading the observation memory  $H$ , and an MLP, each preceded by layer normalization with scale-shift  $(\gamma, \beta)$  and followed by a gated residual connection ( $g$ , marked  $+$ ); the triples  $(\gamma, \beta, g)$  are produced from the virtual time  $\tau$  by a zero-initialized map, so the untrained block is the identity (Peebles and Xie, 2023).

On the encoder side, observation  $i$  is described by a few features — its time  $t_i$ , its transformed value  $w(y_i)$  with  $w$  the learnable monotone map of Sec. 4.3, an index identifying the sensor or state component that produced the value (multivariate states contribute one token per observed component per time), and the design-size feature of Sec. 4.3 — and lifted to the model width by a learned linear map. The attention mechanism that lets these tokens interact is the subject of the next subsection; the choice of features is the subject of Sec. 4.3.

This scheme was not the first attempt but the survivor of a sequence of measured alternatives; the failed ones are retained in the package as reproducible controls and are compared under a common protocol in the ablation study of Appendix B. All sizes are configuration parameters. One default configuration (width 64, four heads, three blocks per side) is used unchanged for every experiment in this paper; at the training-set sizes we use, models of this scale proved sufficient — doubling width and depth changes no benchmark result (Sec. 8.4) — and nothing in the construction caps the scale.

## 4.2 Attention phases from observation times

We adapt rotary position embeddings — the standard positional scheme of modern transformers (Su et al., 2024) — to irregular time sampling.

First the classical construction. Let  $x_1, \dots, x_K \in \mathbb{R}^{d_{\text{model}}}$  be the token states entering an attention layer. Each head computes queries, keys, and values by learned linear maps,

$$q_i = W_Q x_i, \quad k_i = W_K x_i, \quad u_i = W_V x_i, \quad W_Q, W_K, W_V \in \mathbb{R}^{d_h \times d_{\text{model}}}, \quad (5)$$

and token  $i$  aggregates the others with weights  $a_{il} = \text{softmax}_l(\hat{q}_i^\top \hat{k}_l / \sqrt{d_h})$  applied to the  $u_l$ , where  $\hat{q}, \hat{k}$  are position-modulated versions of  $q, k$  defined as follows. Split each query and key into  $d_h/2$  two-dimensional blocks,  $q_i = (q_i^{(1)}, \dots, q_i^{(d_h/2)})$  with  $q_i^{(j)} \in \mathbb{R}^2$ , and rotate block  $j$  by an angle  $\phi_i$  proportional to a per-token scalar:

$$\hat{q}_i^{(j)} = R(\omega_j \phi_i) q_i^{(j)}, \quad \hat{k}_i^{(j)} = R(\omega_j \phi_i) k_i^{(j)}, \quad R(\phi) = \begin{pmatrix} \cos \phi & -\sin \phi \\ \sin \phi & \cos \phi \end{pmatrix}. \quad (6)$$

In the classical scheme the scalar is the token’s integer position in the sequence,  $\phi_i = n_i$ , and the frequencies form a geometric ladder across blocks,  $\omega_j = \omega_{\text{max}}(\omega_{\text{min}}/\omega_{\text{max}})^{(j-1)/(d_h/2-1)}$ , so that high-

frequency blocks resolve neighboring positions and low-frequency blocks long ranges. Because rotations preserve inner products up to the relative angle,

$$\hat{q}_i^\top \hat{k}_l = \sum_{j=1}^{d_h/2} (q_i^{(j)})^\top R(\omega_j(\phi_l - \phi_i)) k_l^{(j)}, \quad (7)$$

every attention logit depends on the positions only through differences  $\phi_l - \phi_i$  — relative position awareness with no positional parameters.

For irregularly sampled data the integer position is meaningless, and the generalization is immediate: we set  $\phi_i = t_i$ , the physical time stamp of observation  $i$ , and choose the frequency range so that the periods  $2\pi/\omega_j$  span from the finest observable time gap to several times the horizon. Nothing else changes; by (7), every logit now depends on the time stamps only through differences  $t_l - t_i$ . The substitution of physical time for position is the same one that time-aware attention models for irregular clinical and event series make in other positional schemes (Horn et al., 2020; Shukla and Marlin, 2021). Two consequences follow at once: the encoder is exactly permutation-invariant (relabeling tokens leaves the right-hand side of (7) unchanged; verified numerically to  $10^{-8}$ ), and variable designs require no positional bookkeeping. The phases enter only the encoder’s self-attention; the cross-attention from parameter slots to the observation memory is unmodified attention, since parameter slots carry no time stamp and the memory states already encode temporal structure.

The adaptation is also motivated by a measured failure. An earlier configuration used integer-position rotations, trained with long token sequences subsampled in place (kept tokens retained their spread-out positions), and evaluated on contiguously packed short sequences. Every component was individually correct, yet the mismatch between the positional distributions seen at training and evaluation produced systematic posterior biases of +1.1 to +3.3 standard deviations (measured in Appendix B). Integer positions create an implicit train/test contract on token placement; physical-time phases eliminate the contract rather than manage it.

### 4.3 Input features

The remaining design choices concern what the tokens contain. Three of them matter and are described in turn: a learnable transform of the observed values, an optional increment-based token construction for directly observed diffusions, and an explicit design-size feature. Each occupies a few lines of code; each is load-bearing, as the ablations of Appendix B show.

**Value transform.** State variables of the benchmark systems live on very different scales, and some — prices, concentrations, populations — are multiplicative: the information sits in relative rather than absolute changes, and the coordinate in which such a variable is well-behaved is logarithmic. Rather than prescribing coordinates per problem, each value enters its token through a learned monotone map,

$$w(x) = \sum_{k=1}^S a_k \operatorname{sign}(x) \log(1 + |x|/s_k), \quad a_k \geq 0. \quad (8)$$

Each basis function is linear for  $|x| \ll s_k$  and logarithmic for  $|x| \gg s_k$ , so the family contains the identity, the signed logarithm, and every compression in between; positivity of the weights makes  $w$  monotone by construction; and — the operative property — every candidate compression is present from initialization, so training only reweights fixed nonlinearities, and the downstream network can exploit whichever coordinate helps from the first optimizer step. The scales  $s_k$  sit on a fixed geometric grid covering the dynamic range of the data ( $S = 8$  octaves by default, one setting for all fifteen systems, no per-problem adjustment).

This structure has to earn its place against simpler-looking learnable alternatives, and we tested them on GBM, where the correct coordinate (log-price) is known and the cost of missing it is a measured +0.48 sd volatility bias. A generic two-layer MLP on the raw values, and a random-Fourier-feature map, both left the bias in place at matched budget: a free-form map must discover the shape and the scale of the compression simultaneously, from gradients arriving through the whole downstream network, and in practice it does not. A classical one-parameter statistical family — the Yeo–Johnson power transforms (Yeo and Johnson, 2000), in essence  $x \mapsto ((1 + |x|)^\lambda - 1)/\lambda$  applied with sign symmetry, which passes through the identity at  $\lambda = 1$  and tends to the logarithm as  $\lambda \rightarrow 0$  — also failed, in an instructive way: trained end-to-end,  $\lambda$  never left its initialization, because a single global exponent must traverse a

gradient plateau before the downstream network has learned anything that would reward the move. The mixture (8) removes the bias reliably, and on GBM the trained weights concentrate on the small-scale components: the model learns the log-price coordinate.

**Increment features for directly observed diffusions.** For scalar SDEs whose state is observed directly — no latent components — the package offers a second token construction: after sorting by time, consecutive observations contribute

$$[t_{(i)}, w(y_{(i)}), \Delta w_i, (\Delta w_i)^2, w_t(\Delta t_{(i)})], \quad \Delta w_i = w(y_{(i+1)}) - w(y_{(i)}).$$

The motivation is elementary: for a Markov observed process the likelihood factorizes over consecutive pairs, transition densities of diffusions depend on increments and gaps, and squared increments carry the quadratic-variation information that identifies diffusion coefficients. This is an inductive bias of the same character as convolutions for images — useful where its assumption holds, not a requirement of the method: the point-token path of Sec. 4.1 is fully general, and the ablations of Appendix B quantify what the increment tokens buy at fixed budget.

The rule for choosing between the constructions — increment tokens exactly when the observed process is Markov — was tested in both directions: an increment-token model class transfers across Markov systems without adjustment (GBM to OU), and on the Heston model, whose observed price is not Markov because volatility is latent, increment and point tokens perform identically across all 25 evaluation cases, as the factorization argument predicts. The package selects the construction from a per-problem flag.

Two further normalization details matter for processes with jumps, where squared increments within a single observation set span several orders of magnitude and a global feature normalization becomes unstable — concretely, on the Merton benchmark the running normalization scale is set by rare jump outliers and the resulting representation drift produces a +1.12 sd volatility bias. First, increment features are logarithmically compressed before normalization; this is a near-identity on systems without jumps (since  $\log(1+x) \approx x$  for small  $x$ ) and by itself turns the Merton failure into a calibrated posterior across all 25 evaluation cases (Sec. 9). Second, the normalization is made adaptive to the observation set: robust summary statistics  $s$  of the observation set, pooled over non-padding positions, are mapped by a zero-initialized MLP to per-channel scale and shift corrections,

$$h_c \mapsto (1 + \gamma_c(s)) h_c + \beta_c(s) \tag{9}$$

— a feature-wise affine modulation, FiLM in the terminology of Perez et al. (2018). Its measured contribution is specific and modest: it removes a residual +0.27 sd bias that survives on the densest Merton designs (the effective jump-detection threshold shifts with the unknown volatility), and it does so as well as a hand-crafted per-observation-set rescaling, within error bars ( $+0.13 \pm 0.06$  vs  $+0.09 \pm 0.05$  sd; Sec. 9). Three implementation conditions are necessary and were each found through a failure: zero initialization (training starts from the working baseline), modulation applied *after* the running normalization (the reverse ordering re-creates the drift it corrects), and pooled statistics that exclude padding positions.

**Design size.** Posterior width is governed by the number of observations — in regular models the error scales as  $K^{-1/2}$  — but  $K$  is a global property of the set, and softmax attention, which computes normalized weighted means, is nearly invariant to it. Softmax attention computes normalized weighted means, which are nearly invariant to the number of tokens — yet posterior width is governed by  $K$ . The deficit is measurable by linear probes from the pooled encoder output: all architecture variants of Table 12 recover posterior-mean statistics ( $R^2 = 0.83\text{--}0.94$ ), while  $R^2(\log K)$  ranges from 0.985 to 0.498 and ranks the variants exactly in the order of their width errors. The repair is to make the count available, and the package default is the simplest form:  $\log K$  enters every token as one of the input features. An alternative that requires no explicit feature — appending a few learned auxiliary tokens to the set, in the spirit of register tokens (Darcet et al., 2024), and letting the model accumulate the count in them — achieves the same effect in our ablations; the number of such tokens is a hyperparameter with no measurable influence beyond one. With the count available, the width error on the hardest case ( $K = 100$ ) drops from 1.95 to 1.49 with bias unchanged; the residual is closed by the training procedure of the next section.

## 5 Training procedures

**Common settings.** Sampling integrates the velocity field with a fixed-step midpoint scheme, 20 steps by default; the step count was fixed once by a convergence check (fourth-order Runge–Kutta with three

times as many steps changes no benchmark statistic), and evaluation uses an exponential moving average of the training weights. All models are trained with Adam at a base learning rate of  $3 \times 10^{-4}$  under a cosine schedule with linear warmup, batch size 64, and gradient clipping; evaluation uses an exponential moving average of the weights with decay 0.999. At every optimizer step the virtual times  $\tau$  are drawn afresh (stratified over  $[0, 1]$ ) together with fresh reference noise  $m_0$ , so no pairing between observation sets and virtual times is ever repeated. For fixed-design problems the training set consists of  $n$  simulated observation sets and training runs for a fixed number of epochs over it; the gallery of Sec. 8.2 uses  $n = 40,000$  simulations and 35 epochs.

**Design amortization.** Training data are simulated once on a fine grid; observation designs are then resampled *from the stored fine trajectories at every optimizer step*. The procedure has three components, each fixed by a controlled comparison on three systems (GBM, Ornstein–Uhlenbeck, FitzHugh–Nagumo):

1. *Fresh designs at every step.* Re-drawing each trajectory’s observation subset at each step makes design memorization impossible and lets one simulation serve arbitrarily many designs. The alternative — one fixed design per trajectory — closes the dense-design error at long budgets but overfits the design set, producing overconfidence at sparse  $K$  (width ratio 0.86 against the exact reference).
2. *The design-size law.*  $K$  is drawn 50% log-uniform on  $[K_{\min}, K_{\max}]$  and 50% uniform on the dense half of the range. Log-uniform sampling alone places  $\approx 6\%$  of the mass at  $K \geq 100$  and starves the dense regime, which appears as inflated widths precisely there; the mixture alone halves the dense-design excess (GBM  $1.27 \rightarrow 1.14$ , OU  $1.34 \rightarrow 1.14$ ; the FitzHugh–Nagumo dense- $K$  SBC failure at  $p < 10^{-3}$  disappears,  $p = 0.54$ ).
3. *Budget scaling.* When higher accuracy is required, simulations and optimizer steps are scaled together at constant epochs; Sec. 8.4 shows the residuals then follow power laws in this single budget variable, and Sec. 8.4 measures why the scaling must be joint: passes over reused observation vectors eventually destroy accuracy, while fresh simulations do not.

With this procedure, the GBM posterior width matches the exact per-design posterior to 5–7% uniformly over  $K = 5, 9, 20, 100$  (Fig. 7).

## 6 Evaluation methodology

Two instruments with different roles are used throughout, plus a diagnostic probe.

### 6.1 Simulation-based calibration as a screen

SBC (Talts et al., 2018) tests a necessary condition: for  $m^* \sim p(m)$  and  $\mathcal{D} \sim p(\mathcal{D} \mid m^*)$ , the rank of each component of  $m^*$  among posterior samples must be uniform. We run SBC per parameter (uniformity test on 500 simulated observation sets with 200 posterior draws each unless stated;  $p > 0.05$  passes) and report coverage of central 50% and 90% intervals.

The power of the test at this size is limited, and because our conclusions depend on knowing that limit, we measured it: on cases where an exact reference is available, SBC at 300–500 observation sets fails to reject posterior width ratios up to  $\approx 1.3$ , and its per-parameter  $p$ -values are unstable between retrainings when residual biases are in the range 0.1–0.2 posterior standard deviations. SBC also cannot penalize a posterior that ignores the data and returns the prior; the prior-mean and ridge controls of Sec. 7 exist for that reason. SBC is therefore used as a cheap screen, never as a verdict.

### 6.2 Reference posteriors as the record

Wherever a tractable likelihood exists, we compare distributions on the same observed points. The headline statistic is a single number per observation set, the squared Fréchet distance between Gaussian fits of the two sample sets in reference-normalized coordinates (each parameter divided by the reference posterior standard deviation),

$$d_F^2 = \|\mu_a - \mu_r\|^2 + \text{tr}(\Sigma_a + \Sigma_r - 2(\Sigma_r^{1/2}\Sigma_a\Sigma_r^{1/2})^{1/2}) \quad (10)$$

— the construction used as the FID score in the image-generation literature, and we adopt that name.



**Why the image-style construction is not enough.** In image generation, FID needs no likelihood because reference samples come straight from the dataset. The analogue here is a *joint-space* FID, which is likelihood-free too: draw  $(m_i, \mathcal{D}_i)$  from the prior and the simulator, draw  $m'_i \sim q(m | \mathcal{D}_i)$  from the trained model, and compare the two sets of pairs  $\{(m_i, \mathcal{D}_i)\}$  and  $\{(m'_i, \mathcal{D}_i)\}$ . We measured this construction on the GBM comparison of Sec. 8.3, where the exact per-design FID is known for every arm, and it fails as an instrument of record in both ways that matter: it compresses a  $200\times$  range of per-design error into  $25\times$  (this package  $0.04 \rightarrow 0.002$ ; the worst external arm  $8.5 \rightarrow 0.03$ ), and it misranks — NPE on raw prices (per-design FID 4.8) scores *better* in joint FID than NPE on hand-derived log-returns (per-design 1.5; joint 0.028 vs 0.045). The mechanism is averaging: Gaussian moments of the pooled joint blend per-observation-set errors, so width inflation dilutes and  $\mathcal{D}$ -correlated bias inflates, in different proportions per arm. A joint two-sample statistic is therefore a necessary-condition screen of the same tier as SBC, and comparisons on fixed observed points remain the record (per-arm joint values in **results/**). FID here is zero for a perfect match, dimensionless, and combines the three ways a posterior can be wrong: location error, width error, and correlation (orientation) error. For scale, a bias of 0.1 reference-sd in each of  $d$  dimensions gives FID  $0.01 d$ , and a posterior 30% too wide in one dimension gives FID  $\approx 0.09$ . Where the *direction* of a failure matters — too wide is merely wasteful, overconfident is silently dangerous — we additionally quote its decomposition per parameter: the signed posterior-mean error in reference-sd units (*bias*) and the ratio of standard deviations (*width*; 1 exact, above one conservative, below one overconfident). References are of three kinds (full details in Appendix C):

- *Closed form.* The linear-Gaussian benchmark (exact Gaussian posterior) and geometric Brownian motion, whose log-returns give a conjugate normal-inverse- $\chi^2$  posterior importance-corrected to the box prior.
- *Exact-likelihood MCMC.* An adaptive random-walk Metropolis sampler (after Haario et al., 2001): proposal scale adapted to a 0.25 acceptance target during burn-in, proposal shape replaced by the empirical covariance Cholesky factor rescaled to  $2.38/\sqrt{d}$ , adaptation frozen before the retained phase, out-of-box proposals rejected (which samples the box-truncated target exactly). Likelihoods are exact per-gap transition densities: Gaussian Euler–Maruyama chain transitions for OU (matching the simulator that generated the data, plus the stationary density of the informative initial state), a Poisson-mixture transition density for the Merton jump-diffusion, log-normal residuals around the Bateman curve for pharmacokinetics, and a Gaussian observation model around the deterministic PDE solve for Fisher–KPP — in the last case one likelihood evaluation is one PDE solve.
- *Chain validation.* A reference is used only after validation. Every MCMC reference in this report is run at least twice from well-separated starting points, and the between-chain discrepancy of posterior means (in posterior-sd units) is reported with the result it certifies. Where a closed form and a chain both exist, the chains reproduce it to a worst mean discrepancy of 0.099 sd and width ratios 0.91–1.03 over 24 comparisons.

Twice during development the screen and the record contradicted each other; both times the contradiction exposed an implementation error (a padding mask leaking into set-level statistics; the positional contract of Sec. 4.2). We consider the redundancy a feature of the methodology rather than an inefficiency.

### 6.3 Encoder probing

To separate encoder failures from flow failures, we fit ridge regressions from the pooled encoder output to known sufficient statistics of the problem. High probe  $R^2$  with poor posterior accuracy localizes the defect downstream of extraction. In the Merton analysis of Sec. 9, the failing configuration extracted robust volatility statistics *better* than the healthy one ( $R^2 = 0.967$  on truncated realized variance), redirecting the repair from the encoder’s capacity to the stability of its input representation.

## 7 Benchmark suite

The suite contains fifteen systems (Table 1), defined below with their equations, the meaning and prior range of every parameter, and the observation model. Variable-design training — observation subsets resampled at every optimizer step — is the package default and the recommended mode: beyond enabling inference at any design, the resampling acts as data augmentation over designs. Nine systems

| system                     | type                | $d$ | obs.          | design   | reference posterior   |
|----------------------------|---------------------|-----|---------------|----------|-----------------------|
| linear-Gaussian            | linear map          | 4   | full          | fixed    | exact Gaussian        |
| Ornstein–Uhlenbeck         | SDE                 | 2   | full          | fixed    | exact-likelihood MCMC |
| geometric Brownian motion  | SDE                 | 2   | full          | fixed    | conjugate closed form |
| Cox–Ingersoll–Ross         | SDE                 | 3   | full          | fixed    | — (Euler pseudo-MLE)  |
| double well                | SDE                 | 3   | full          | fixed    | —                     |
| stochastic Lotka–Volterra  | 2-d SDE             | 4   | full          | fixed    | —                     |
| FitzHugh–Nagumo            | 2-d SDE             | 4   | $v$ only      | fixed    | —                     |
| SEIR                       | 4-d SDE             | 5   | $I, D$        | fixed    | —                     |
| polynomial-drift SDE       | SDE                 | 5   | full          | fixed    | —                     |
| Heston                     | 2-d SDE             | 5   | price only    | variable | —                     |
| Merton jump-diffusion      | jump SDE            | 5   | full          | variable | Poisson-mixture MCMC  |
| Hénon–Heiles               | Hamiltonian ODE     | 3   | $q_1$ only    | variable | —                     |
| Hodgkin–Huxley             | 4-d ODE             | 4   | $V$ only      | variable | —                     |
| pharmacokinetics (Bateman) | ODE (closed form)   | 3   | concentration | variable | log-normal MCMC       |
| Fisher–KPP                 | PDE (64-point grid) | 2   | 3 sensors     | variable | PDE-likelihood MCMC   |

**Table 1:** The benchmark suite. “obs.” lists the observed coordinate when the state is only partially visible; “design” distinguishes the fixed-grid gallery from variable-design systems, where every observation set carries its own set of (time, sensor) observation points. Dashes in the last column indicate systems where no tractable reference exists — these are evaluated by SBC and controls only, which is precisely the regime that motivates carrying exact-reference systems in the same suite.

are nevertheless benchmarked on conventional fixed grids, for two practical reasons: their closed-form or cheap references live on those grids, and the external tools of Sec. 8.3 consume fixed-length inputs.

All SDEs are integrated by Euler–Maruyama on the stated fine grid; ODEs by RK4 (Hodgkin–Huxley by a semi-implicit scheme for the gating variables); the PDE by finite differences. Systems marked *random designs* are trained and evaluated in the design-amortized mode of Sec. 2: the observation times (and, for the PDE, sensor indices) of every observation set are drawn at random, and one trained network serves all of them; the remaining systems use one conventional fixed grid. The *classical estimator* listed for each system is the standard non-Bayesian method a practitioner would apply; all of them return point estimates, not distributions, and they serve as an accuracy floor for the posterior mean. Three recur and are defined here once. *Kramers–Moyal regression* estimates the drift and squared diffusion from the conditional first and second moments of the observed increments ( $f(x) \approx \mathbb{E}[\Delta X \mid X = x]/\Delta t$ ,  $\sigma^2(x) \approx \mathbb{E}[\Delta X^2 \mid X = x]/\Delta t$ ) and fits the parametric form to these estimates by least squares. *Multi-start nonlinear least squares* fits the deterministic solve of the system to the observed trajectory by minimizing squared error over parameters, restarted from many random initial points to escape local minima. *Exact maximum likelihood* is available where the transition density is known in closed form. Fixed-design observations are token sets over the union of the stated channels; “fast/slow” channels observe every step over a short window and every  $j$ -th step over the horizon respectively. Priors are uniform on the stated boxes. Classical estimators (`sota()`) are listed per system.

**Linear-Gaussian.**  $y = Am + \varepsilon$ ,  $\varepsilon \sim \mathcal{N}(0, 0.5^2 I_6)$ , with a fixed correlated design matrix  $A \in \mathbb{R}^{6 \times 4}$ ;  $m \in [-3, 3]^4$ . Exact Gaussian posterior; the classical estimator is the posterior mean itself. This system checks the flow engine in isolation from the encoder.

**Ornstein–Uhlenbeck.**  $dX = -\theta X dt + \sigma dW$ , stationary start  $X_0 \sim \mathcal{N}(0, \sigma^2/2\theta)$ ;  $dt = 0.02$ , 500 steps; channels: fast ( $1 \times 50$ ), slow ( $20 \times 24$ ). Prior  $\theta \in [0.3, 3]$ ,  $\sigma \in [0.2, 1.5]$ . Classical: exact MLE from per-gap Gaussian transitions.

**Geometric Brownian motion.**  $dS = \mu S dt + \sigma S dW$ ,  $S_0 = 1$ ;  $dt = 0.01$ , 500 steps (five “years”); channels: fast ( $1 \times 60$ ), slow ( $10 \times 40$ ). Prior  $\mu \in [-0.2, 0.4]$ ,  $\sigma \in [0.1, 0.6]$ . Classical: exact MLE on log-returns. Reference: conjugate posterior (Appendix C).

**Cox–Ingersoll–Ross.**  $dX = a(b - X) dt + \sigma\sqrt{X} dW$ ;  $dt = 0.02$ , 500 steps; channels as OU. Prior  $a \in [0.3, 3]$ ,  $b \in [0.3, 1.5]$ ,  $\sigma \in [0.1, 0.5]$ . Classical: Euler pseudo-likelihood MLE (Cox et al., 1985).

**Double well.**  $dX = (\theta_1 X - \theta_2 X^3) dt + \sigma dW$ ;  $dt = 0.01$ , 1000 steps, noise strong enough for barrier crossings; channels: fast ( $1 \times 80$ ), slow ( $25 \times 36$ ). Prior  $\theta_{1,2} \in [0.5, 3]$ ,  $\sigma \in [0.4, 1.2]$ . Classical: Kramers–Moyal regression (defined above).

**Stochastic Lotka–Volterra.** Prey/predator pair with multiplicative noise,  $dx_1 = (\alpha x_1 - \beta x_1 x_2) dt + 0.1 x_1 dW_1$ ,  $dx_2 = (\delta x_1 x_2 - \gamma x_2) dt + 0.1 x_2 dW_2$ ; both components observed;  $dt = 0.01$ , 600 steps; channels: fast ( $1 \times 50$ ), slow ( $15 \times 40$ ). Prior  $\alpha, \gamma \in [0.8, 1.5]$ ,  $\beta, \delta \in [0.4, 1.2]$ . Classical: multi-start nonlinear least squares (defined above).

**FitzHugh–Nagumo.**  $\dot{v} = v - v^3/3 - w + I + \text{noise}$ ,  $\dot{w} = \epsilon(v + a - bw)$ ; only  $v$  observed at 25 evenly spaced times over horizon 40 ( $dt = 0.05$ ), observation noise 0.05. Prior  $a, b \in [0.5, 0.9]$ ,  $\epsilon \in [0.05, 0.3]$ ,  $I \in [0, 0.5]$ . Classical: multi-start nonlinear least squares (defined above).

**SEIR with mortality.** Five-compartment stochastic epidemic (two-phase contact rate  $\beta_1 \rightarrow \beta_2$ , incubation  $\alpha$ , recovery  $\gamma_r$ , mortality  $\gamma_d$ ); infected and deceased compartments observed at 20 days over a 60-day horizon ( $dt = 0.1$ ), observation noise 0.004. Prior  $\beta_1 \in [0.2, 0.6]$ ,  $\beta_2 \in [0.05, 0.3]$ ,  $\alpha \in [0.2, 0.5]$ ,  $\gamma_r \in [0.05, 0.2]$ ,  $\gamma_d \in [0.005, 0.05]$ . Classical: nonlinear least squares on  $(I, D)$ .

**Polynomial-drift SDE.**  $dX = (c_0 + c_1 X + c_2 X^2 + c_3 X^3) dt + \sigma dW$  — drift identification on a dense polynomial basis, the setting of SINDy (Brunton et al., 2016) without its sparsity prior (a sparse-coefficient variant is discussed in Sec. 10);  $dt = 0.01$ , 1000 steps; channels: fast ( $1 \times 80$ ), slow ( $20 \times 45$ ). Prior  $c_0, c_2 \in [-0.5, 0.5]$ ,  $c_1 \in [-2, 0]$ ,  $c_3 \in [-1, -0.1]$ ,  $\sigma \in [0.3, 0.8]$ . Classical: least squares of the Kramers–Moyal drift estimate on the polynomial basis — the dense-basis version of SINDy regression.

**Heston (random designs).**  $dS = \mu S dt + \sqrt{v} S dW_1$ ,  $dv = \kappa(\bar{v} - v) dt + \xi \sqrt{v} dW_2$ ,  $\text{corr}(dW_1, dW_2) = \rho$  (Heston, 1993); price observed, volatility latent;  $dt = 0.01$ , 500 steps, designs of  $K \in [2, 128]$  random times. Prior  $\mu \in [-0.1, 0.3]$ ,  $\kappa \in [0.5, 5]$ ,  $\bar{v} \in [0.02, 0.3]$ ,  $\xi \in [0.1, 1]$ ,  $\rho \in [-0.9, 0]$ .

**Merton jump-diffusion (random designs).**  $dS/S = \mu dt + \sigma dW + d(\sum_i (e^{J_i} - 1))$  with Poisson jump rate  $\lambda$  and  $J_i \sim \mathcal{N}(\mu_J, s_J^2)$  (Merton, 1976);  $dt = 0.01$ , 500 steps,  $K \in [2, 128]$ . Prior  $\mu \in [-0.1, 0.3]$ ,  $\sigma \in [0.1, 0.5]$ ,  $\lambda \in [0.2, 3]$ ,  $\mu_J \in [-0.3, 0.3]$ ,  $s_J \in [0.05, 0.4]$ . Reference: Poisson-mixture transition density (Appendix C).

**Hénon–Heiles (random designs).** Two-dimensional Hamiltonian oscillator with the coupling potential used in Lubich et al. (2015), frequencies  $(\omega_1, \omega_2)$  and coupling  $\lambda$ ;  $q_1$  observed at random times with noise;  $dt = 0.05$ , 800 steps,  $K \in [4, 64]$ . Prior  $\omega_{1,2} \in [0.7, 1.3]$ ,  $\lambda \in [0.02, 0.22]$ .

**Hodgkin–Huxley (random designs).** Standard four-dimensional membrane model (Hodgkin and Huxley, 1952); voltage observed at random times; parameters  $(g_{Na}, g_K, g_L, I)$  with prior  $[60, 180] \times [10, 50] \times [0.05, 0.5] \times [4, 12]$ ; 3000 integration steps,  $K \in [4, 96]$ .

**Pharmacokinetics (random designs).** The standard one-compartment oral-dosing model of pharmacokinetics (e.g. Gabrielsson and Weiner, 2000): absorption at rate  $k_a$ , elimination at rate  $k_e$ , distribution volume  $V$ , giving the Bateman concentration curve, concentration  $c(t) = \frac{D_0 k_a}{V(k_a - k_e)} (e^{-k_e t} - e^{-k_a t})$  with dose  $D_0 = 500$ ; log-normal assay noise (10%); horizon 24 h,  $K \in [3, 64]$  irregular draws. Prior  $k_a \in [0.5, 4]$ ,  $k_e \in [0.05, 0.4]$ ,  $V \in [20, 100]$ . Reference: log-normal likelihood MCMC.

**Fisher–KPP (random designs).**  $\partial_t \theta = D \partial_{xx} \theta + r \theta(1 - \theta)$  on  $[0, 1]$  with reflecting boundaries, 64-point finite-difference grid, Gaussian initial bump; three fixed sensors at  $x = 16/63, 32/63, 48/63$  observed at random (time, sensor) pairs with noise 0.02;  $dt = 0.01$ , 200 steps,  $K \in [4, 96]$ . Prior  $D \in [0.001, 0.01]$ ,  $r \in [2, 10]$ . Reference: Gaussian observation model around the deterministic solve; one likelihood evaluation is one PDE solve.

|                                        | requires                    | train     | per obs. set  |
|----------------------------------------|-----------------------------|-----------|---------------|
| exact conjugate posterior (GBM)        | closed-form factorization   | —         | <1 ms         |
| exact-likelihood MCMC (GBM, OU)        | tractable likelihood        | —         | 42–124 ms     |
| mixture-likelihood MCMC (Merton, PK)   | tractable likelihood        | —         | 0.03–0.1 s    |
| PDE-likelihood MCMC (Fisher–KPP)       | forward solver in the chain | —         | 271 s         |
| <b>sbi</b> NPE / FMPE                  | simulator only              | 1–4 min   | 19–703 ms     |
| BayesFlow 2 (coupling / flow matching) | simulator only              | 13–21 min | 0.3–97 ms     |
| Simformer port                         | simulator only              | 10–12 min | $\approx$ 1 s |
| <b>amortix</b> (this package)          | simulator only              | 4–63 min  | 70 ms         |

**Table 2:** References and baselines. Top block: reference posterior samplers, the instruments of record; cost is per observation set for 2,000 samples on a laptop CPU (chain validation, where used, doubles it; the Fisher–KPP entry is  $\approx$ 12,000 PDE solves — the number of likelihood evaluations any MCMC spends per observation set whatever the forward model costs). Bottom block: amortized estimators, trained once per system; external tools at the 20,000-simulation comparison budget of Sec. 8.3, **amortix** over the full range used in this report (20,000–480,000 simulations). Amortized inference is per observation set for 2,000 samples, one H200 GPU.

**Controls.** Two model-free controls accompany every result. The *prior-mean predictor* ignores the data entirely (exactly 25% mean absolute error under a uniform prior). The *ridge control* is a point predictor of the parameters: a degree-2 ridge regression on generic summary statistics of the token set — per token feature, the mean, standard deviation, mean square, mean absolute value, and the 0.1/0.5/0.9 quantiles, computed over however many observations are present. It recovers no distribution and is not meant to; its role is to detect a specific silent failure that the calibration screen cannot see: a posterior that ignores the data and returns the prior passes SBC perfectly, and only a point-accuracy control unmasks it.

## 8 Experimental results

This section is organized as an experimental study: the protocol — references, baselines, model sizes, budgets — is fixed first; the main results (posterior overlays and the global metric on every system) follow; then the comparison with existing tools, the scaling behavior in budget, model size and data reuse, and the evaluation of design amortization.

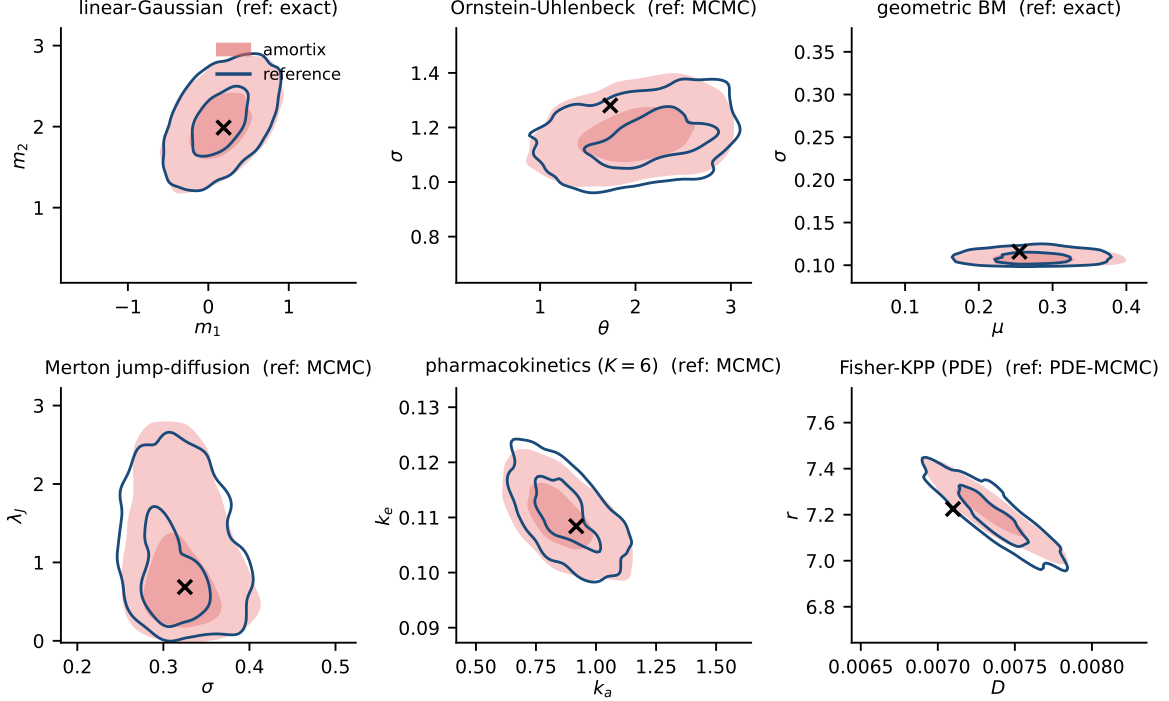
### 8.1 Experimental protocol

**References and baselines.** Every accuracy number in this section is the FID of Sec. 6.2 against a reference posterior computed on the same observed points, reported as the median over the system’s test battery (the distribution over observation sets is right-skewed; means are quoted where the tail is the finding, and for scale, a 0.1 sd bias in each of  $d$  dimensions gives FID 0.01  $d$ ). Table 2 lists the reference samplers — what each requires and what it costs — and the external amortized tools of the comparison. Simulation-based calibration is reported as the number of parameters passing the uniformity screen at  $p > 0.05$  over 500 simulated observation sets (*SBC pass*); with 32 such tests on the fixed-grid systems, one or two rejections are expected by chance even from an exact posterior.

**Models and training.** For each reference system we train the same architecture at three sizes — tiny (0.1M parameters), the default small (0.5M), and big (2.3M) — plus two miniature sizes (nano, 26k; pico, 7k) used to locate the capacity floor. All runs follow the procedure of Sec. 5, with one coupling that must be respected: at width 128 the base learning rate is halved, otherwise optimizer instability masquerades as a capacity effect (Sec. 8.4). Simulation budgets range from 5,000 to 480,000 per system. The main-result tables use each system’s best trained model; the scaling study (Sec. 8.4) reports the budget  $\times$  size grid behind that choice. Observation designs are random at training and at evaluation wherever the system defines a design distribution; fixed designs appear only where the interface of an external tool or reference requires them (Sec. 8.3, linear-Gaussian check).

### 8.2 Main results

Figure 2 shows the object the package produces — the joint posterior itself. For every system with an exact or validated reference, the posterior of the trained model is overlaid on the reference for one



**Figure 2:** Joint posteriors of `amortix` (filled contours: 39% and 86% highest-density regions) against exact or validated-MCMC reference posteriors (open contours, same levels) on one test observation set per system; crosses mark the true parameters. Top row: systems with closed-form or cheap exact references. Bottom row: systems where the reference itself required exact-likelihood MCMC with two-chain validation — including the ridge-shaped Fisher-KPP posterior, where the model reproduces the curvature, not just the moments.

representative test observation set — three systems with simple posteriors that serve as checks (linear-Gaussian, GBM, OU), and three whose posterior geometry is nontrivial: the Merton jump-diffusion, where volatility and jump rate compete to explain the same variance; pharmacokinetics at six irregular blood draws, the data-poor clinical regime; and Fisher-KPP, whose posterior concentrates on the curved ridge  $Dr \approx \text{const}$  fixed by the observable front speed. The learned posteriors reproduce location, scale, orientation, and — where present — the curvature of the ridge.

Table 3 reports the corresponding global metric on the full test batteries, and Table 4 the calibration screen for the nine systems where no tractable reference exists. Every reference row is evaluated in the package’s default mode — one network per system, a fresh random design per test observation set — and reads median FID 0.01–0.07 on four of the six systems. The two elevated entries are measured effects rather than noise: Merton (0.19) is limited by model capacity at this simulation budget (Sec. 8.4), and pharmacokinetics (0.18) is the one system whose posterior contracts below the amortized precision floor at dense designs (Sec. 8.5). On the screen systems, 34 of 36 parameters pass, with the marginal cases moving between retrainings as expected at the screen’s resolution limit. Per-parameter  $p$ -values, rank distributions, and coverage curves are in Appendix A.

Pharmacokinetics shows the clearest budget effect in the suite — at a third of the saturation budget its FID is an order of magnitude worse — and its third test set illustrates a failure mode of the metric rather than of the model: the true volume of distribution lies within 3% of the prior boundary and the reference posterior for  $k_e$  has a standard deviation of 1% of its value, so a 0.02 absolute shift is magnified into  $\text{FID} \approx 38$ ; behavior at prior boundaries is listed as a limitation in Sec. 10. Together with Table 3, this figure is the central evidence that one small default model recovers full posteriors across the suite.

### 8.3 Comparison with existing tools

**Protocol.** We compare against the directly comparable amortized estimators available in maintained packages: NPE with its default masked autoregressive flow (Papamakarios et al., 2017) and FMPE (Wildberger et al., 2023) from `sbi` 0.27 (Boelts et al., 2024); the coupling-flow and flow-matching networks of BayesFlow 2.0 (Radev et al., 2020); and, because it is the closest architecture to ours in the literature,

| system                    | $d$ | reference              | random designs, fresh per set | FID  |
|---------------------------|-----|------------------------|-------------------------------|------|
| geometric Brownian motion | 2   | conjugate, per design  | $K = 2\text{--}100$ (600)     | 0.01 |
| linear-Gaussian           | 4   | exact subset posterior | $K = 2\text{--}6$ (300)       | 0.03 |
| Ornstein–Uhlenbeck        | 2   | exact-likelihood MCMC  | $K = 2\text{--}100$ (600)     | 0.05 |
| Fisher–KPP (PDE)          | 2   | PDE-likelihood MCMC    | $K = 10\text{--}90$ (96)      | 0.07 |
| pharmacokinetics          | 3   | log-normal MCMC        | $K = 5\text{--}50$ (96)       | 0.18 |
| Merton jump-diffusion     | 5   | Poisson-mixture MCMC   | $K = 5\text{--}100$ (96)      | 0.19 |

**Table 3:** Systems with reference posteriors: pooled median FID (10) of the trained design-amortized model of the default architecture. Every row is evaluated in the package’s default mode: each test observation set (count in parentheses) draws a fresh random design of the stated size range, and the reference posterior is recomputed on exactly those points. Per- $K$  breakdown: Table 10. The two elevated entries have measured mechanisms: pharmacokinetics meets the amortized precision floor at dense designs (Sec. 8.5), and Merton is capacity-limited at this simulation budget (Sec. 8.4).

| system                    | $d$ | SBC pass |     |
|---------------------------|-----|----------|-----|
|                           |     | small    | big |
| Cox–Ingersoll–Ross        | 3   | 3/3      | 1/3 |
| double well               | 3   | 3/3      | 2/3 |
| stochastic Lotka–Volterra | 4   | 3/4      | 4/4 |
| FitzHugh–Nagumo           | 4   | 4/4      | 3/4 |
| SEIR                      | 5   | 5/5      | 5/5 |
| polynomial-drift SDE      | 5   | 5/5      | 4/5 |
| Heston                    | 5   | 5/5      | 5/5 |
| Hénon–Heiles              | 3   | 3/3      | 1/3 |
| Hodgkin–Huxley            | 4   | 3/4      | 3/4 |

**Table 4:** Systems without a tractable reference: parameters passing the SBC uniformity screen ( $p > 0.05$ , 500 observation sets  $\times$  200 draws; random-design systems over mixed designs), for the same saturated small and big models as Table 3. Individual marginal cases move between configurations and retrainings, as expected at the screen’s resolution limit (Sec. 6); at the same data budget and hyperparameters the big model passes fewer (28/36 vs 34/36).

a port of the Simformer (Gloeckler et al., 2024) built from its authors’ minimal example (transformer–diffusion over the joint, random condition masks, their architecture and SDE settings, full 75,000-step training). All arms receive the same simulation budget (20,000 draws from the same prior and simulator), run with their package defaults and their own stopping rules, and are compared on exactly three axes: training cost (wall-clock time to fit at this budget), accuracy (posterior-mean bias and width ratio per parameter against the reference posterior, as in Sec. 6.2), and inference cost (wall-clock time per observation set for 2,000 posterior samples). Non-amortized inference sets the scale on both ends: on GBM, whose likelihood is a three-line formula, adaptive Metropolis costs 42 ms per observation set and nothing amortized is worth its training bill for a single instance; on Fisher–KPP, one likelihood evaluation is one PDE solve, so the same reference sampler spends  $\approx 12,000$  solves — minutes of computation — on *each* observation set; and for Heston or Hodgkin–Huxley no likelihood formula exists at any price. Because these estimators consume fixed-length vectors, the comparison uses fixed observation designs — the only experiments in this report, apart from the linear-Gaussian check, run in that mode: a fixed design is the interface the external tools require, not the mode this package is built for (Sec. 8.5). Three systems are used, chosen to separate the possible explanations of any performance gap: a multiplicative process where input conditioning is hostile (GBM, exact conjugate reference), an additive process where it is comparatively benign (OU, exact-likelihood MCMC reference, two validated chains per observation set), and a PDE where every likelihood evaluation requires a forward solve (Fisher–KPP), so that the cost relationship between reference and amortized inference is inverted. Where informative, external arms are additionally given hand-derived inputs — log-prices, or per-gap log-returns, the exact coordinates of the GBM likelihood factorization — to separate the effect of the input representation from the generative engine.

**Multiplicative process (GBM).** Table 5 compares all methods on one device and one metric. At package defaults on raw price data, every external configuration misses the posterior by one to two

| method                    | input                  | train  | inference   | FID       |
|---------------------------|------------------------|--------|-------------|-----------|
| <b>amortix</b> CFM        | raw prices             | 236 s  | 70 ms       | 0.04      |
| <b>sbi</b> NPE            | raw prices             | 170 s  | 36 ms       | 4.8       |
| <b>sbi</b> FMPE           | raw prices             | 215 s  | 359 ms      | 3.8       |
| BayesFlow 2 coupling      | raw prices             | 1252 s | 0.3 ms      | 2.6       |
| BayesFlow 2 flow matching | raw prices             | 790 s  | 95 ms       | 2.3       |
| Simformer port            | raw prices             | 703 s  | 1049 ms     | 8.5       |
| <b>sbi</b> NPE            | log-prices             | 157 s  | 25 ms       | 5.8       |
| <b>sbi</b> NPE            | log-returns            | 52 s   | 26 ms       | 1.5       |
| <b>sbi</b> FMPE           | log-returns            | 99 s   | 378 ms      | 0.8       |
| BayesFlow 2 coupling      | log-returns            | 1254 s | 0.3 ms      | 1.2       |
| BayesFlow 2 flow matching | log-returns            | 771 s  | 97 ms       | 1.4       |
| Simformer port            | log-returns            | 586 s  | 1053 ms     | 0.34      |
| exact-likelihood MCMC     | log-returns (analytic) | —      | 42 ms (CPU) | reference |

**Table 5:** GBM: all methods at package defaults, one H200 GPU, equal 20,000-simulation budget, 200 test observation sets against the exact conjugate posterior. FID is the Fréchet distance (10) to the reference, median over test observation sets (the distribution across sets is right-skewed; means are 1.5–3 $\times$  larger, per-arm values in **results**/). Middle block: raw data, the amortized default regime. Bottom block: the same external tools given hand-derived input coordinates. Inference time is per observation set for 2,000 samples.

| method (raw values)   | train | inference    | FID       |
|-----------------------|-------|--------------|-----------|
| <b>amortix</b> CFM    | 226 s | 77 ms        | 0.07      |
| <b>sbi</b> NPE        | 134 s | 19 ms        | 2.7       |
| <b>sbi</b> FMPE       | 114 s | 279 ms       | 1.2       |
| exact-likelihood MCMC | —     | 124 ms (CPU) | reference |

**Table 6:** Ornstein–Uhlenbeck (additive control), fixed 74-point design, 20,000-simulation budget, 100 test observation sets, one H200 GPU. FID: median Fréchet distance to the exact-likelihood MCMC reference (two validated chains per observation set).

orders of magnitude more than **amortix**: FID 2.3–4.8 against 0.04. The dominant mechanism is the input representation, not the generative engine (Sec. 9); **sbi** itself emits a warning about precision loss during z-scoring on this data. Hand-derived coordinates (bottom block of the table) help but do not converge to a common answer: with identical log-return inputs the external tools land at median FID 0.34–1.5 — the best of them still an order of magnitude above **amortix** operating on raw data — and they fail in different directions: NPE remains over-dispersed ( $\sigma$  width 1.44), both BayesFlow arms become overconfident (widths 0.42–0.85, the silently dangerous direction), and the Simformer port, best by median, has the heaviest tail across observation sets (mean FID 1.9, five times its median). Log-prices alone do not help NPE at all (5.8 vs 4.8 raw). At fixed input coordinates the flow-matching engine beats the autoregressive flow (0.8 vs 1.5 on log-returns), consistent with the engine choice of Sec. 3. Every defect here is invisible without an exact reference and lies within or near the blind region of SBC at practical sizes (Sec. 6); the failure directions replicate across devices and seeds (per-parameter decompositions for an independent laptop-CPU run of every arm are in **results**/).

**Additive control (Ornstein–Uhlenbeck).** OU is an additive process, so the floating-scale problem of raw prices is absent by construction, and running the identical protocol on it shows how much of the GBM failure that problem accounts for (Table 6). The external tools improve substantially: their volatility bias becomes consistent with zero ( $+0.19 \pm 0.19$  sd for NPE,  $+0.08 \pm 0.12$  for FMPE), and their width inflation drops from 2.5–2.6 to 1.9 (NPE) and 1.6 (FMPE). It does not disappear: extracting quadratic-variation information from 74 raw values remains a representation task that a z-scored vector input leaves entirely to the network. **amortix** reads 1.03 and 1.05 on the same observation sets.

**Simulator-limited regime (Fisher–KPP).** The third system inverts the cost structure of the first two: every likelihood evaluation requires solving the forward model. A one-dimensional reaction–diffusion equation is inexpensive in absolute terms (16 ms per solve here); what makes the case informative is the *ratio* between reference and amortized inference, because both sides scale linearly with the solve cost

| method (40 observed values) | train | inference   | FID       |
|-----------------------------|-------|-------------|-----------|
| <b>amortix</b> CFM          | 591 s | 69 ms       | 0.07      |
| <b>sbi</b> NPE              | 235 s | 19 ms       | 0.5       |
| <b>sbi</b> FMPE             | 163 s | 703 ms      | 76        |
| PDE-likelihood MCMC         | —     | 271 s (CPU) | reference |

**Table 7:** Fisher–KPP, fixed  $K = 40$  spatio-temporal design, 20,000-simulation budget, 32 test observation sets, one H200 GPU. FID: median Fréchet distance to the PDE-likelihood MCMC reference ( $\approx 12,000$  PDE solves per observation set; the amortized model’s entire training equals the reference cost of fewer than three observation sets). The FMPE value is its diffusivity failure, discussed in the text; the defect is invisible to calibration screens at practical sizes.

— a chain of  $n$  likelihood evaluations costs  $n$  solves per observation set whatever the forward model, while the amortized cost per observation set does not involve the simulator at all. The measured ratio below transfers to genuinely expensive simulators, where the same reference sampler would take days per observation set. A single fixed design of  $K = 40$  random (time, sensor) points is shared by all observation sets and methods; the reference is the PDE-likelihood sampler of Sec. 6.2 with two validated chains per observation set (271 s per observation set for the pair; two-chain validation over 32 observation sets: worst mean discrepancy 0.22 sd, mean 0.07 sd). Table 7 gives the results. **amortix** matches the reference at FID 0.07, answering in 69 ms per observation set, and its entire training cost (591 s) equals the reference cost of fewer than three observation sets — amortization pays for itself almost immediately in any repeated-inference setting. NPE shows moderate error (FID 0.5, from widths of 1.26–1.28); FMPE at defaults fails outright on the diffusivity ( $+7.3 \pm 2.7$  sd bias at  $12.5\times$  width, FID 76) while remaining reasonable on the reaction rate — default configurations behave discontinuously across parameters, detectable here only because the expensive reference was computed.

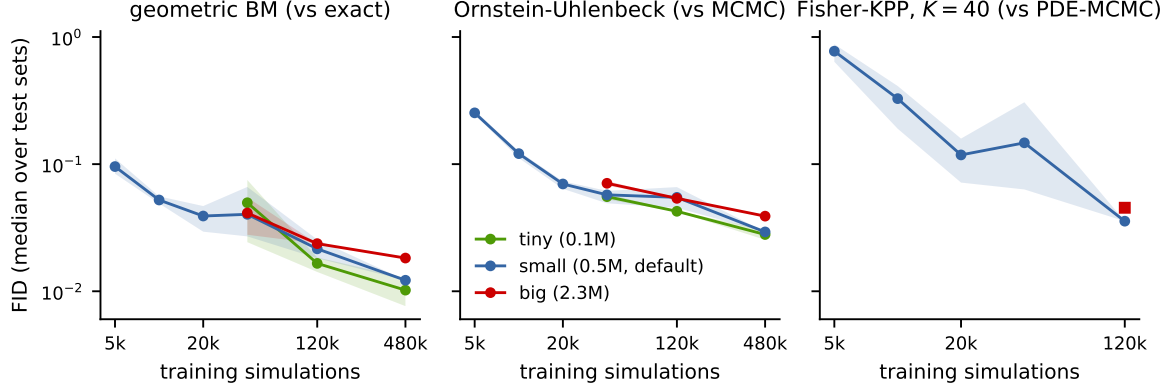
**Scope of the comparison.** Every external arm ran at its package defaults under our simulation budget; specialists could improve each with problem-specific embedding networks or summary statistics. The comparison is designed to measure exactly that dependence: what accuracy is delivered when the input representation is not hand-engineered per problem. The Simformer arm in particular is a port of a minimal example rather than a tuned configuration, and its numbers should be read as representative of that recipe, not of the method’s ceiling. Conversely, the variable-design capability of Sec. 8.5 is not exercised here at all, since the external estimators consume fixed-length inputs by default.

## 8.4 Scaling: budget, model size, and data reuse

Accuracy is reported over the budget  $\times$  size grid of the protocol rather than at a single operating point. Figure 3 shows median FID for the three model sizes over simulation budgets from 5,000 to 480,000 (optimizer steps scale with the budget at constant epochs; the big model runs at its width-scaled learning rate). Three observations. First, FID falls monotonically with budget on every system across two orders of magnitude — GBM 0.10  $\rightarrow$  0.010, OU 0.25  $\rightarrow$  0.028, Fisher–KPP 0.77  $\rightarrow$  0.036 — and the data axis did not saturate anywhere we measured. Second, at  $d = 2$  the three sizes are interchangeable: tiny, small, and big coincide within seed spread along the entire budget range, so capacity is not the axis that buys accuracy on low-dimensional systems — Table 9 below extends this to five sizes and locates where it stops being true. Third, the widening seed band on Fisher–KPP at intermediate budgets is informative rather than noise: the residual error lies along the weakly identified ridge direction  $\ln(r/D)$ , which contributes little to the loss (3), so independently trained models retain seed-level freedom along it; under the variable-design procedure, which averages every optimizer step over fresh designs, the same ridge error decays steadily. As a control, refining the sampling ODE solver from 20 midpoint steps to 60 RK4 steps changes nothing measurable.

**Fresh simulations beat repeated passes.** Table 8 varies the number of simulated trajectories and the number of epochs independently, for the fixed-design GBM and OU models at the default size. Two readings. Along constant compute — 10,000 trajectories  $\times$  120 epochs versus  $120,000 \times 10$ , equal optimizer steps — fresh data wins by an order of magnitude or more in FID. And passes are not merely diminishing but destructive: at 120 epochs every configuration is worse than at 35, including the largest training set (OU: 0.066  $\rightarrow$  0.648). The mechanism is memorization of the tokens: virtual times and reference noise are redrawn at every step (Sec. 5), so what repeats across passes is only the fixed observation vectors.





**Figure 3:** Median FID versus simulation budget at three model sizes (both axes logarithmic; optimizer steps scale with the budget at constant epochs; the big model uses the width-scaled learning rate). Lines: seed means; bands: seed ranges where multiple seeds were run. On the two-parameter systems (left, middle) the three sizes coincide within seed spread — the budget axis, not the parameter axis, buys accuracy. Fisher-KPP (right): default size, with the big model at 120,000 marked by the square. References as in Table 2.

| system | trajectories | FID at epochs |       |       |                   |
|--------|--------------|---------------|-------|-------|-------------------|
|        |              | 10            | 35    | 120   | 120 + subsampling |
| GBM    | 10,000       | 0.105         | 0.059 | 0.965 | 0.114             |
|        | 30,000       | 0.036         | 0.047 | 0.659 | —                 |
|        | 120,000      | 0.044         | 0.026 | 0.131 | 0.277             |
| OU     | 10,000       | 0.206         | 0.151 | 1.892 | 0.253             |
|        | 30,000       | 0.090         | 0.095 | 1.648 | —                 |
|        | 120,000      | 0.038         | 0.066 | 0.648 | 0.143             |

**Table 8:** Median FID when the same fixed-design training is given more simulated trajectories (rows) or more passes over them (columns), at the default model size; cells on an anti-diagonal cost equal compute. The last column repeats the 120-epoch run with a fresh random 75% token subset at every optimizer step.

Consistent with this, redrawing a random 75% token subset at every optimizer step (`token_dropout`) largely repairs the 120-epoch column — OU  $1.89 \rightarrow 0.25$  at 10,000 trajectories,  $0.65 \rightarrow 0.14$  at 120,000 — while slightly hurting the one cell that had barely overfitted (GBM at 120,000:  $0.13 \rightarrow 0.28$ ): training on subsets while evaluating on the full design is its own train/test mismatch, so subsampling is a repair, not a free improvement. The design-amortized procedure performs the same augmentation natively and without the mismatch — every optimizer step sees a fresh design drawn from the law used at evaluation — which is why the amortized network of Table 10 is at least as accurate as its fixed-design counterpart at the same budget.

**The capacity floor moves with the parameter dimension.** Table 9 trains the same architecture at five sizes spanning 7 thousand to 2.3 million parameters, at a fixed budget of 120,000 simulations, on four systems of increasing parameter dimension. At  $d = 2$  the ladder is flat: a 7,000-parameter model,  $70\times$  smaller than the default, matches the default within seed noise on GBM. At  $d = 3$  the floor is visible: below  $\sim 10^5$  parameters accuracy degrades several-fold. In both places the floor announces itself first as seed instability rather than as a shift of the median — OU at the smallest size spans 0.05–0.10 across seeds and pharmacokinetics 0.33–0.57, while the better-provisioned rows reproduce to a few thousandths. At  $d = 5$  the ladder is monotone through its entire range — every halving of width costs accuracy, and this is the one column where the largest model is not worse than the default — so the Merton entry of Table 3 is capacity-limited, not data-limited. Two couplings matter when reading such ladders. The learning rate must shrink with width: at width 128 the default  $3 \times 10^{-4}$  is unstable and masquerades as “bigger is worse” (GBM median FID 0.068), while halving it restores parity (0.024); the big row of the table reports the stable setting. And adding parameters without adding data buys nothing here, whereas adding data does: at 480,000 simulations the default model reaches FID 0.012 (GBM) and 0.029 (OU), while the big model at the same enlarged budget and rescaled learning rate reads 0.018 and 0.039. The

| model (width $\times$ blocks)    | params    | GBM<br>$d = 2$ | OU<br>$d = 2$ | pharmacokin.<br>$d = 3$ | Merton<br>$d = 5$ |
|----------------------------------|-----------|----------------|---------------|-------------------------|-------------------|
| pico ( $8 \times 2$ )            | 7,114     | 0.028–0.029    | 0.052–0.095   | 0.33–0.57               | 0.48              |
| nano ( $16 \times 2$ )           | 25,730    | 0.021–0.027    | 0.042–0.054   | 0.26–0.29               | 0.30              |
| tiny ( $32 \times 2$ )           | 97,522    | 0.014–0.018    | 0.042–0.043   | 0.10–0.12               | 0.23–0.25         |
| small ( $64 \times 3$ , default) | 483,346   | 0.018–0.026    | 0.045–0.066   | 0.055                   | 0.21              |
| big ( $128 \times 4$ )           | 2,321,554 | 0.024          | 0.054         | 0.085                   | 0.21              |

**Table 9:** Capacity ladder: median FID at a fixed budget of 120,000 simulations; sizes are given as width  $\times$  transformer blocks, with parameter counts for the GBM instance (counts vary by a few percent across systems with the input dimension). Ranges span independent training seeds. GBM and OU use the fixed-design protocol of Table 8; pharmacokinetics and Merton use the design-amortized protocol and the evaluation batteries of Table 3. The big row uses the width-rescaled learning rate (see text). The ladder is flat at  $d = 2$ , develops a floor near  $10^5$  parameters at  $d = 3$ , and is monotone through the full range at  $d = 5$ .

same lever closes the capacity gap at  $d = 5$ : quadrupling the simulations improves the default Merton model from 0.213 to 0.186 and the big one from 0.206 to 0.200, so once data is scaled the default *overtakes* the big model — capacity binds at a fixed simulation budget, but the data axis ultimately outbids the parameter axis everywhere we measured. The one exception is pharmacokinetics, where the enlarged budget does not improve the battery median ( $0.055 \rightarrow 0.098$  on its 32-set battery) — consistent with a model operating at the amortized precision floor (Sec. 8.5), where seed-level jitter of order the floor itself dominates any battery this size. On low-dimensional systems the parameter axis saturates by a few thousand weights; the simulation axis did not saturate on any other system.

Costs, for orientation. The saturated models of Tables 3 and 4 train in 19–63 minutes per system on one H200 GPU (120,000 simulations; 66–72k optimizer steps); at the 20,000-simulation budget of the comparisons above, training takes 6–12 minutes. On a laptop CPU, training takes roughly four times longer and produces bit-identical posteriors, because all stochastic generation is kept on a CPU random number generator by design. Inference draws 2,000 posterior samples in 426 ms per observation set on the laptop CPU and 72 ms batched (135 ms single) on the GPU. The small external models of Sec. 8.3 train and sample faster at their defaults; the tiny-model regime is kernel-launch-bound on GPUs (sbi’s default flow trains  $6\times$  slower there than on the laptop CPU).

## 8.5 Amortization over observation designs

The package’s central claim is that one trained network answers  $p(m \mid S)$  for *every* observation design  $S$  — any number of points, any placement. Table 10 and Fig. 4 test the claim on every system of the suite where a per-design reference can be computed and validated — six of the fifteen: each test observation set draws a fresh random design, the reference posterior is recomputed on exactly those points, and FID is evaluated per design-size bucket. Two readings. First, wherever a fixed-design specialist of the same architecture and simulation budget exists, the single amortized network is comparable to it at the specialist’s own design size (GBM 0.018 vs 0.012; OU 0.023 vs 0.029; linear-Gaussian 0.046 vs 0.039; Fisher–KPP 0.050 vs 0.036) and is the only one of the two that answers anywhere else on the  $K$  axis. Amortization over designs is not a trade-off against accuracy at any point we measured; it also removes a failure mode, since resampling the observation subset at every optimizer step doubles as data augmentation (Sec. 8.4 measures what the fixed-design procedure pays for reusing identical tokens across passes). Second, the two systems with elevated FID are not noise, and the two paragraphs after the table take them apart.

**The amortized precision floor.** Pharmacokinetics is the one system whose FID *rises* with the design size ( $0.116 \rightarrow 0.421$  from  $K = 5$  to 50), and we audited the number before interpreting it: an independent exact posterior (Laplace mode plus importance sampling with  $2 \times 10^5$  exact likelihood evaluations per observation set) agrees with the MCMC reference to 0.0–0.23 sd, and the model’s FID against either reference coincides — the value is real, not a reference artifact. It is also not a budget artifact: a model trained on  $4\times$  the simulations and  $3\times$  the optimizer steps is no better on the same test sets. The mechanism is a race. FID is measured in units of the reference posterior’s width, and that width contracts with  $K$ : for the smooth Bateman curve with 10% assay noise, the  $k_e$  posterior at  $K = 50$  has a standard deviation of 0.5–0.9% of the prior range. The network’s *absolute* error stops contracting at roughly  $10^{-3}$  of the prior range (back-computed from the measured FID) — a floor that neither capacity nor data moved in any experiment of Sec. 8.4 — and once the posterior is narrower than a few multiples of that

| system                    | FID at design size $K$ (median) |            |             | specialist     |
|---------------------------|---------------------------------|------------|-------------|----------------|
|                           | sparse                          | medium     | dense       | at its own $K$ |
| geometric Brownian motion | 0.006 (5)                       | 0.008 (20) | 0.018 (100) | 0.012 (95)     |
| Ornstein–Uhlenbeck        | 0.072 (5)                       | 0.045 (20) | 0.023 (100) | 0.029 (74)     |
| linear-Gaussian           | 0.012 (2)                       | 0.035 (4)  | 0.046 (6)   | 0.039 (6)      |
| Fisher–KPP                | 0.073 (10)                      | 0.050 (40) | 0.126 (90)  | 0.036 (40)     |
| pharmacokinetics          | 0.116 (5)                       | 0.157 (20) | 0.421 (50)  | —              |
| Merton jump-diffusion     | 0.169 (5)                       | 0.206 (20) | 0.187 (100) | —              |

**Table 10:** One design-amortized network per system, evaluated at design sizes never fixed in advance (the bucket’s  $K$  in parentheses): each test observation set draws a fresh random design and its reference posterior is recomputed on exactly those points. Last column: a separate network of the same architecture and simulation budget trained on a single fixed design, which exists only at its own  $K$ . GBM and OU are evaluated on the denser  $K$ -grid of Fig. 4. Pharmacokinetics rises with  $K$  — the precision floor, audited in the text; Merton is flat in  $K$  and capacity-limited (Sec. 8.4).

floor, relative error grows mechanically. The stochastic systems never reach this regime: diffusion noise keeps the GBM and OU posteriors near 5% of the prior range even at  $K = 100$ , which is why their FID falls with  $K$ ; Fisher–KPP touches the floor only at its densest bucket (0.126 at  $K = 90$ ).

**Where references end.** Two further systems — Hénon–Heiles and Hodgkin–Huxley — are deterministic flows with Gaussian observation noise, so their likelihood is tractable in principle and a per-design reference looks within reach. It is not, and the reason is worth recording. Under deterministic chaos the posterior concentrates on needle-thin sets (Hénon–Heiles at  $K = 20$ : the  $\omega_1$  posterior has a standard deviation near  $10^{-3}$  of the prior range while  $\omega_2$  stays prior-wide), and adaptive random-walk chains do not converge on such geometry: independent chains land 67–108 sd apart on half of the Hénon–Heiles test sets, and 1.8–18 sd apart on Hodgkin–Huxley, against the 0.22 sd worst case of every validated reference in this report. By the validation standard of Sec. 6.2 these references are rejected, and both systems remain under the calibration screen of Table 4. The precision-floor argument above predicts that a valid reference, if computed by a stronger sampler, would find elevated FID here as well.

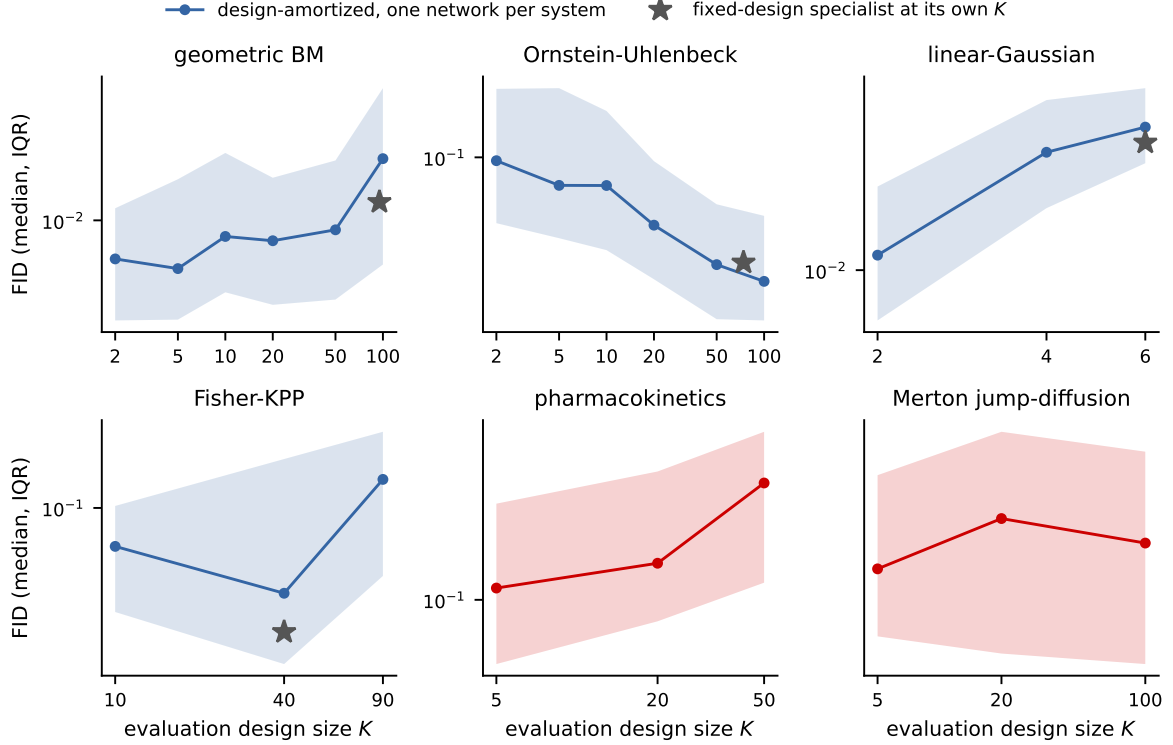
**Variable-design zoo.** With the class rule and training procedure fixed, six further systems were trained without per-system adjustments. Heston (five parameters, latent volatility): all 25 SBC cases pass, and increment tokens give no advantage over point tokens, consistent with the class boundary (Sec. 4.3). Merton jump-diffusion: 25 of 25 cases after the heavy-tail treatment of Sec. 9, including dense-design cases that failed at  $p < 10^{-3}$  before it; the residual width sits on the budget curve below. Hénon–Heiles (parameters of the potential of Lubich et al. 2015, observing one coordinate at random times): calibrated at every design size down to  $K = 6$ , where the posterior widens honestly rather than degrading. Hodgkin–Huxley (Hodgkin and Huxley, 1952): calibrated on mixed designs; one dense-design case shows the budget signature. Pharmacokinetics: at  $K = 6$  irregular blood draws, bias  $-0.01$  to  $-0.06$  sd and widths 1.09–1.33 against exact-likelihood MCMC. Fisher–KPP: the posterior concentrates on the ridge  $D \cdot r \approx \text{const}$ ; the residual along-ridge bias decays with budget (Sec. 8.4).

Which encoder components make design amortization work is an architecture question, separate from the accuracy claims above; the controlled comparison of eight encoder variants that fixed the design of Sec. 4 is in Appendix B.

## 9 Failure analysis: conditioning of the encoder input

Every accuracy failure encountered during the development of this package — including, we argue, the external failures of Table 5 — reduces to a single cause appearing in three forms: the conditioning of the numbers presented to the set encoder. We document the three forms with their measurements and repairs; all three repairs are learnable modules rather than problem-specific transformations.

**Form 1: floating multiplicative scale.** On GBM, raw price tokens force the network to divide squared increments by squared levels internally; with global standardization this manifests as a volatility bias of  $+0.48$  posterior sd (confirmed against the closed-form posterior) that no amount of training removes. The repair is the monotone mixture reparametrization of Sec. 4.3, which learns the log coordinate;



**Figure 4:** FID as a function of the evaluation design size, for one design-amortized network per system: median (line) and interquartile range (band) over the test observation sets of each bucket (100 per bucket for GBM, OU, and linear-Gaussian, 32 for the MCMC-referenced systems), each set with a fresh random design and a per-design reference. Stars: fixed-design networks of the same architecture and simulation budget, which exist only at their own  $K$ . Both axes logarithmic; note the different vertical scales.

the Yeo–Johnson ablation there shows why the parametrization, not the hypothesis class, is the operative choice. External estimators exhibit the same form on the same data (Table 5, raw-price rows).

**Form 2: cardinality blindness.** Normalized attention erases set size, which controls posterior width under design amortization; the probe measurements and the  $\log K$ /register-token repairs are given in Sec. 4.3, with the arm-by-arm consequences in Table 12.

**Form 3: heavy tails.** On the Merton jump-diffusion, squared increments have sample standard deviation  $3.1 \times 10^4$  against a diffusion signal of order  $10^{-2}$ –1; running feature normalization never converges, and the flow trains against a drifting representation, producing a +1.12 sd volatility bias. Encoder probing (Sec. 6.3) shows the information is extracted — the failing configuration encodes robust volatility statistics better than the healthy one — so the repair targets representation stability: logarithmic compression of increment features (near-identity on clean diffusions since  $\log(1+x) \approx x$  for small  $x$ ) restores all 25 evaluation cases. A precision follow-up against the Poisson-mixture reference then isolated a smaller (+0.27 sd) residual whose mechanism is a jump-classification threshold that floats with the unknown volatility; a per-observation-set set-conditioning module (pooled statistics modulating feature channels, initialized to the identity) removes it as effectively as a hand-crafted per-observation-set rescaling (+0.127  $\pm$  0.055 vs +0.094  $\pm$  0.054 sd, statistically indistinguishable). This was one of three such paired comparisons in the project (log coordinate, set size, robust scaling); in all three the learnable module matched the hand-crafted repair once initialized at the working configuration. We summarize the pattern as a design principle: the hand-crafted fix identifies the mechanism and the initialization; the learnable module placed there is what ships.

Three implementation conditions proved necessary for the set-conditioning module and are recorded here because each was found through a failure: the modulation must be zero-initialized (training must start from the working baseline); it must act after feature normalization, not before (the reverse ordering re-creates the representation drift it is meant to cure); and pooled set statistics must exclude padding

| system                    | per-parameter SBC $p$ -values |                    |                   |                   |                   |  |
|---------------------------|-------------------------------|--------------------|-------------------|-------------------|-------------------|--|
| linear-Gaussian           | $m_1$ : 0.06                  | $m_2$ : 0.60       | $m_3$ : 0.38      | $m_4$ : 0.56      |                   |  |
| Ornstein–Uhlenbeck        | $\theta$ : 0.28               | $\sigma$ : 0.77    |                   |                   |                   |  |
| SEIR                      | $\beta_1$ : 0.33              | $\beta_2$ : 0.22   | $\alpha$ : 0.61   | $\gamma_r$ : 0.34 | $\gamma_d$ : 0.53 |  |
| geometric Brownian motion | $\mu$ : 0.26                  | $\sigma$ : 0.44    |                   |                   |                   |  |
| Cox–Ingersoll–Ross        | $a$ : 0.65                    | $b$ : <b>0.001</b> | $\sigma$ : 0.73   |                   |                   |  |
| double well               | $\theta_1$ : 0.78             | $\theta_2$ : 0.54  | $\sigma$ : 0.88   |                   |                   |  |
| stochastic Lotka–Volterra | $\alpha$ : 0.07               | $\beta$ : 0.69     | $\delta$ : 0.08   | $\gamma$ : 0.35   |                   |  |
| FitzHugh–Nagumo           | $a$ : 0.19                    | $b$ : 0.15         | $\epsilon$ : 0.90 | $I$ : 0.63        |                   |  |
| polynomial-drift SDE      | $c_0$ : 0.06                  | $c_1$ : 0.96       | $c_2$ : 0.46      | $c_3$ : 0.80      | $\sigma$ : 0.50   |  |

**Table 11:** Per-parameter SBC uniformity  $p$ -values for the fixed-design gallery run of the standard-budget gallery run (500 observation sets  $\times$  200 draws;  $p > 0.05$  passes). The single failure is marked in bold; see Sec. 8.2 for its behavior across retrainings.

positions (a masking error here made padded and packed representations of the same observation set inequivalent, detected only because SBC and the exact-reference probe disagreed).

## 10 Discussion and limitations

The evidence assembled here supports a specific claim: for dynamical-systems inverse problems, the input representation of an amortized posterior estimator is not a preprocessing detail but the dominant accuracy factor, and it can be made part of the model rather than part of the problem specification. The comparison of Sec. 8.3 shows the cost of the alternative — fixed representations with global standardization — across four generative engines, on a problem where the correct answer is known in closed form. The OU control separates the mechanisms: on additive data the bias of the external estimators disappears and their width inflation drops by roughly a factor of two but does not vanish, so the conditioning pathology accounts for the dominant part of the failure and a smaller representation-learning gap accounts for the rest.

Limitations. (i) The remaining inaccuracies of the suite decay as measured power laws in training compute (Fig. 3); we have not searched for methods that improve the exponents. (ii) The power limits of SBC imply that small residuals can only be certified where an exact or validated reference exists; systems without tractable likelihoods inherit the weaker screen, and carrying reference-equipped systems alongside them in one suite is our current mitigation. (iii) The comparison of Sec. 8.3 uses package defaults; it measures the out-of-the-box regime, not the ceiling of any method. (iv) Bayesian experimental design — the intended consumer of design amortization — is not yet demonstrated; the training procedure of Sec. 5 was built so that the required object  $p(m \mid S)$  is already coherent across subsets  $S$ . (v) No real-data study with posterior predictive checks is included yet. (vi) The polynomial-drift system uses a dense coefficient prior; the scientifically interesting variant is a *sparse* prior — train on coefficient vectors with random supports and test whether the posterior recovers the support — which requires extending the normalization (2) to priors with atoms at zero, a planned extension of the package. (vii) All priors are uniform on boxes; the probit normalization (2) extends to any factorized prior with tractable CDFs, but correlated priors would require a different normalization.

## A Calibration detail for the fixed-grid systems

Per-parameter SBC  $p$ -values for the standard-budget gallery run (40,000 simulations, the run of Figs. 5 and 6) are listed in Table 11; Fig. 5 shows empirical-vs-nominal coverage of the central credible intervals, and Fig. 6 the underlying rank distributions.

## B Encoder ablations for design amortization

The ablations of this appendix track the posterior width component rather than FID: the mechanism under study — set-size blindness of normalized attention — expresses itself specifically as width error, and FID would mix it with unrelated deviations. Table 12 and Fig. 7 compare encoder variants for design amortization on GBM, trained at  $K \sim \log\text{-uniform}[2, 128]$  random-time designs and evaluated against the exact per-design posterior (96 observation sets per bucket). Three observations organize the table. First, index-based positions combined with in-place subsampling during training (leaving kept tokens at

| encoder variant                                   | $\sigma$ width ratio at $K$ |       |       |       | $R^2(\log K)$ |
|---------------------------------------------------|-----------------------------|-------|-------|-------|---------------|
|                                                   | 5                           | 9     | 20    | 100   |               |
| point tokens, index positions, spread subsampling | 0.94*                       | 1.13* | 1.42* | 2.08* | —             |
| point tokens, index positions, contiguous         | 1.05                        | 1.17  | 1.13  | 1.73  | 0.957         |
| point tokens, time-kernel attention bias          | 1.03                        | 1.14  | 1.31  | 2.51  | 0.712         |
| point tokens, edge-message attention              | 1.03                        | 1.15  | 1.29  | 2.45  | 0.498         |
| point tokens, time phases                         | 1.01                        | 1.10  | 1.14  | 1.95  | —             |
| + explicit $\log K$ feature                       | 1.01                        | 1.05  | 1.08  | 1.49  | —             |
| pair tokens, time phases                          | 1.02                        | 1.02  | 1.02  | 1.27  | 0.985         |
| pair tokens + full training procedure             | 1.06                        | 1.06  | 1.05  | 1.07  | —             |

**Table 12:** GBM with variable designs: posterior width for the volatility  $\sigma$  against the exact per-design posterior (96 observation sets per design-size bucket) and, where measured, the probe  $R^2$  for  $\log K$  from the pooled encoder output. The table tracks  $\sigma$  because it is the parameter whose posterior width is governed by the observation density — the drift  $\mu$  is weakly informed at every  $K$ , and all variants recover its posterior within a few percent, so it does not discriminate between architectures. \*This row also carries biases of +1.09 to +3.30 posterior sd (the positional train/test mismatch of Sec. 4.2); all other rows have biases consistent with zero.

spread positions while inference packs contiguously) produce biases of +1.1 to +3.3 posterior sd with no individually incorrect component — the positional pathology of Sec. 4.2; repacking or replacing positions with time phases removes the bias entirely. Second, all point-token variants share inflated widths at dense designs, in proportion to how poorly their pooled representation encodes  $\log K$  (rightmost column; the probe of Sec. 4.3) — including two attention mechanisms specifically constructed to exploit continuous time (a learned per-head time-kernel logit bias in the spirit of relative-position methods, and edge-message attention carrying gap-modulated increments on all pairs). Third, pair tokens with time-phase attention, plus the training procedure of Sec. 5, bring every bucket to within 7% of the exact width.

## C Reference posteriors

**GBM conjugate posterior.** With observations at indices  $i_0 = 0 < i_1 < \dots$  of the fine grid, log-returns  $r_k = \log(S_{i_{k+1}}/S_{i_k})$  over gaps  $\tau_k$  are independent  $\mathcal{N}((\mu - \sigma^2/2)\tau_k, \sigma^2\tau_k)$ . In the coordinates  $(b, \sigma^2) = (\mu - \sigma^2/2, \sigma^2)$  the model is a Gaussian location–scale family with sufficient statistics  $T = \sum_k \tau_k$ ,  $R_1 = \sum_k r_k$ , and  $SS_w = \sum_k r_k^2/\tau_k - R_1^2/T$ ; draws from the standard conjugate posterior ( $\sigma^2 = SS_w/\chi_{n-1}^2$ ,  $b \mid \sigma^2 \sim \mathcal{N}(R_1/T, \sigma^2/T)$ ) are transformed to  $(\mu, \sigma)$  and importance-resampled to the uniform box prior with the appropriate Jacobian weight. The sampler was validated against exact-likelihood MCMC on the same designs: over 24 comparisons, worst posterior-mean discrepancy 0.099 sd, width ratios 0.91–1.03.

**Adaptive Metropolis.** Random-walk Metropolis on the box, proposal  $x' = x + sC\xi$ ,  $\xi \sim \mathcal{N}(0, I)$ . During burn-in (default  $\max(500, n/2)$  iterations) the scalar  $s$  is adapted toward a 0.25 acceptance rate in blocks of 50 with a decaying Robbins–Monro gain, and once sufficient burn-in states are collected the shape  $C$  is replaced by the Cholesky factor of the empirical covariance rescaled by  $2.38/\sqrt{d}$  (Haario et al., 2001); adaptation is frozen for the retained phase. Out-of-box proposals are rejected, which samples the box-truncated target exactly since the proposal is symmetric. Likelihoods: per-gap Euler–Maruyama Gaussian transitions for OU (matching the generative process, plus the stationary density of  $X_0$ ); the exact GBM log-return density; the Poisson-mixture transition density for Merton (truncated at the numerically negligible jump count); log-normal residuals around the Bateman curve; a Gaussian observation model around the deterministic PDE solve for Fisher–KPP. Every chain used as a reference is paired with a second chain from a well-separated start; between-chain posterior-mean discrepancies are reported alongside the results they certify (Sec. 8.3).

## D Correspondence with package options

| in this report                           | in the package                                                     |
|------------------------------------------|--------------------------------------------------------------------|
| point tokens                             | <code>embed="wpoint"</code>                                        |
| pair tokens + set conditioning           | <code>embed="wfilm"</code>                                         |
| warped values/increments (fixed designs) | <code>embed="wbasis" / "wdiff"</code>                              |
| attention phases from physical time      | <code>rope="time"</code>                                           |
| fresh designs each optimizer step        | <code>fit(retokenize=...) via make_retokenizer()</code>            |
| variable-design problem wrapper          | <code>amortix.designs.DesignProblem</code>                         |
| benchmark systems                        | <code>amortix.problems (DESIGN_Z00)</code>                         |
| calibration screen / reference machinery | <code>sbc_design / amortix.mcmc + likelihood factories</code>      |
| comparison scripts (Sec. 8.3)            | <code>examples/baseline_{npe,ou,kpp,bayesflow,simformer}.py</code> |

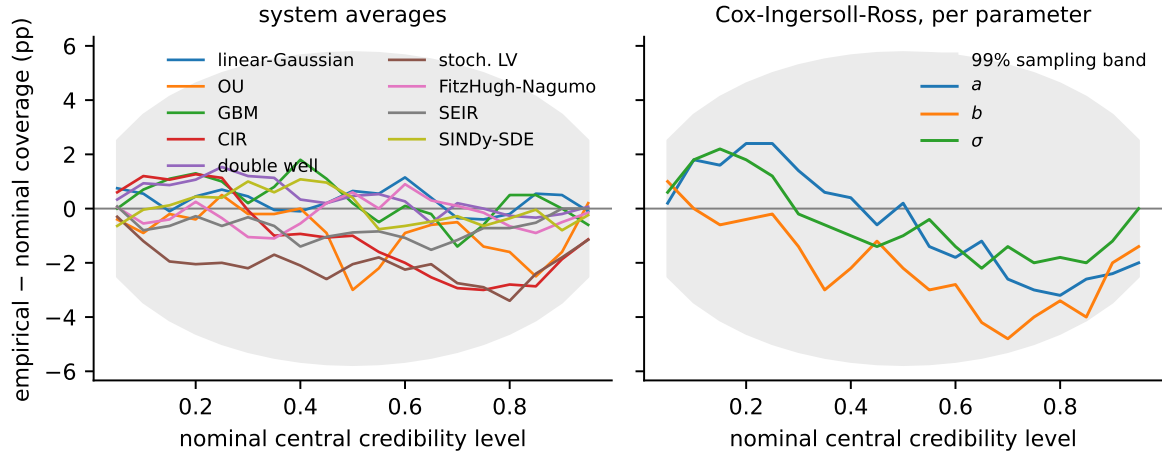
All defaults named in this report are selected automatically by `FlowPosterior(problem)`; the non-default options exist as reproducible controls for the comparisons above.

## References

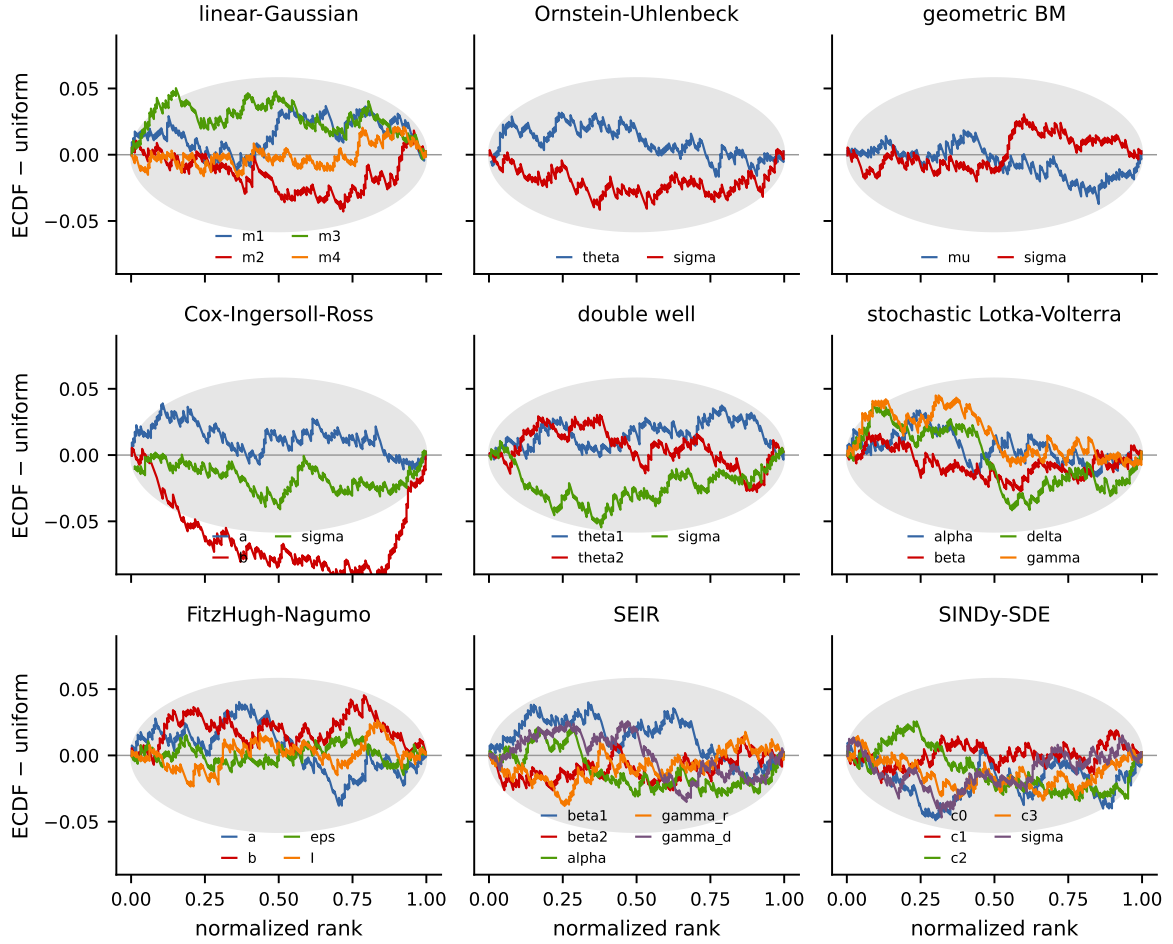
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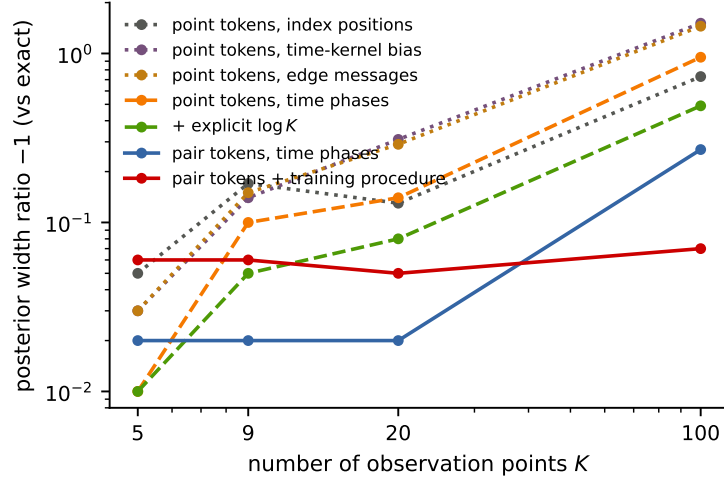




**Figure 5:** Empirical coverage of central credible intervals for the fixed-design gallery, shown as the deviation from the nominal level. For each system, 500 synthetic observation sets are simulated with known parameters and the posterior’s central  $q\%$  interval is checked against the truth; a calibrated posterior has zero deviation at every level, and the gray band is the 99% pointwise range induced by the finite number of test observation sets. Left: deviation averaged over each system’s parameters. Right: individual parameters of Cox–Ingersoll–Ross, the system with the largest deviation in this run; the  $b$  curve reflects the  $\approx 0.15$  sd location shift discussed in the text. Rank-level detail for all 32 parameters is in Appendix A (Fig. 6).



**Figure 6:** Calibration check for the fixed-design gallery. Procedure: for each system, 500 synthetic observation sets are simulated with known parameters; for each observation set the model produces 200 posterior draws, and the rank of the true parameter among the draws is recorded. If the posterior is calibrated, these ranks are uniform. Each curve shows, for one parameter, the deviation of the empirical CDF of normalized ranks from the uniform diagonal; the gray band is the 99% range attributable to sampling noise at 500 observation sets. A curve bulging below zero on the left indicates a posterior shifted low relative to the truth, and vice versa. All curves stay within the band except Cox–Ingersoll–Ross  $b$ , whose deviation corresponds to the  $\approx 0.15$  sd location shift discussed in the text.



**Figure 7:** The data of Table 12 in graphical form: excess posterior width for the GBM volatility — the width ratio against the exact posterior minus one, so 0.1 means a posterior 10% too wide and lower is better — as a function of the number of observation points  $K$  (both axes logarithmic). Dotted curves: point-token variants with no set-size information, which degrade as designs become dense; dashed: partial repairs; solid: increment tokens, and increment tokens combined with the training procedure of Sec. 5, which is flat in  $K$ .